MSP Report - Sprint 2

Franka Traupe, Roschanak Babai, Niklas Sander, Valeriia Baida, Guilherme Fernandes

Github (Sprint 1): `https://github.com/niklassander/MSP/releases/tag/Sprint1_Submission`
Github (Sprint 2): `https://github.com/niklassander/MSP/releases/tag/Sprint2_Submission`

Roles

Scrum Master: Franka Traupe
Product Owner: Guilherme Fernandes
Quality Controllers and Development Team: Franka Traupe, Guilherme Fernandes, Roschanak Babai, Valeriia Baida, Niklas Sander

Features

(Implemented features highlighted in bold)

Basic Features (Implemented in prototype)

- **Register account**
- **Register type of vehicle**
- **Define route**
- **Alert if speed limit is violated**
- **Calculate estimated time of arrival**

Features from Existing System

- **Bookmarking (home, work, custom)**
- Avoid tolls
- **Recent destinations**
- Show current speed (implemented already as part of speed alerts)
- Parking lots
- **Hazard reports**
- **Change units**

New Features

- **Statistics: usage and driving quality (distance traveled, active days etc.)**
- Walking route
- Shared ride

Project Planning: Sprint 1

Task Table

Task ID	Task	Duration (days)	Depends on
1	Explore original app & choose features	3	-
2	Review chosen features	1	1
3	Develop BPMNs for each feature	4	2
4	Review BPMNs	1	3
5	Front-end development	7	4
6	Back-end development	7	4
7	Integrate front-end and back-end	2	5,6
8	Testing	2	7
9	Bug Fixing	1	8

Table 1: Task List with Dependencies

Gantt Chart

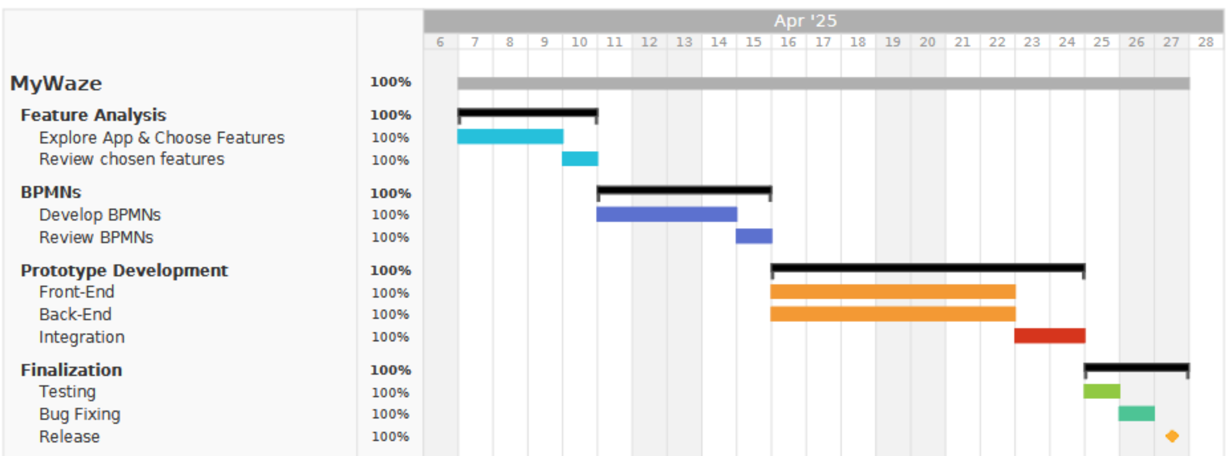


Figure 1: Gantt Chart

Burndown Chart

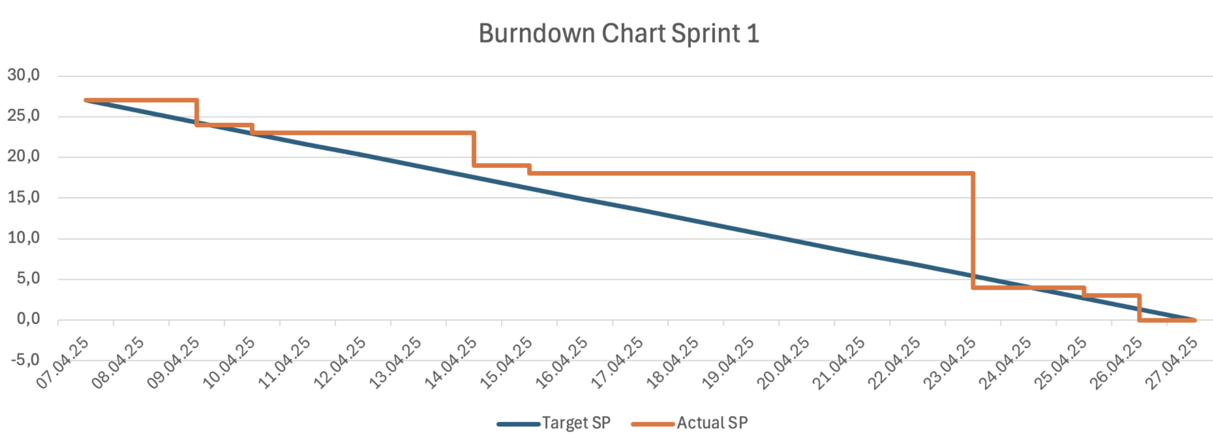


Figure 2: Burndown Chart

Project Planning: Sprint 2

Task Table

Task ID	Task	Duration (days)	Depends on
1	Sprint 1 Revision	2	-
2	Context Diagram	3	1
3	Use Case View	3	1
4	Logical View	3	1
5	Development View	4	4
6	Process View	11	5
7	Prototyping	11	5
8	Deployment View	3	7

Table 2: Task List with Dependencies

Gantt Chart

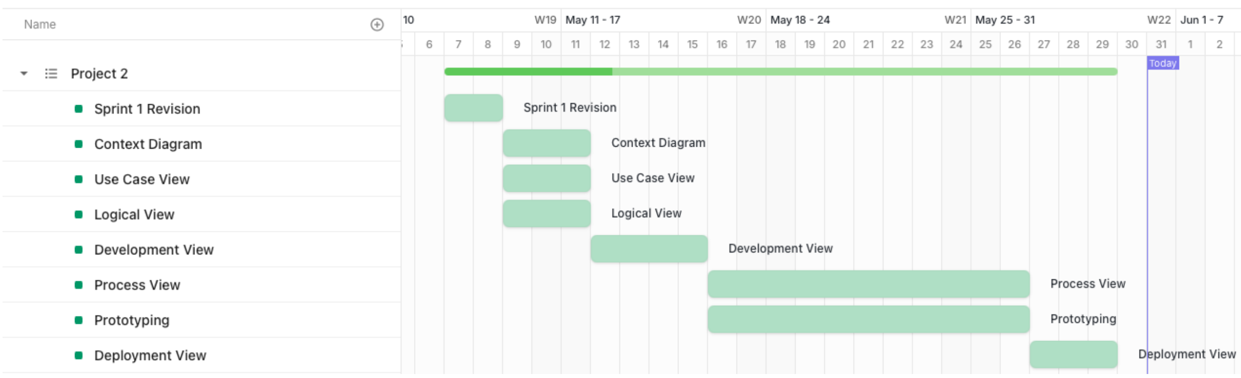


Figure 3: Gantt Chart

Burndown Chart

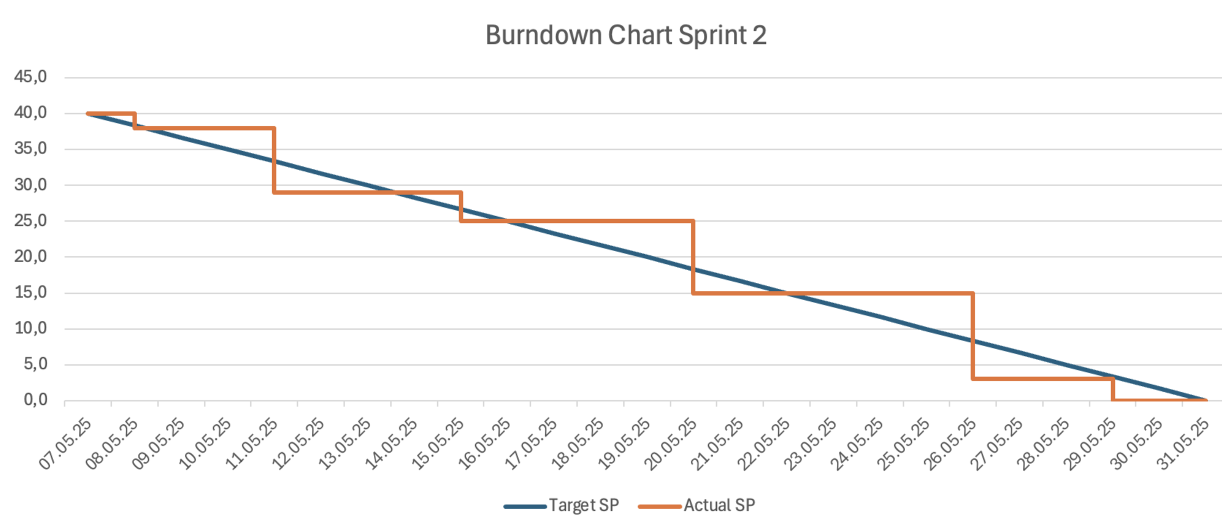


Figure 4: Burndown Chart

Requirements Diagrams

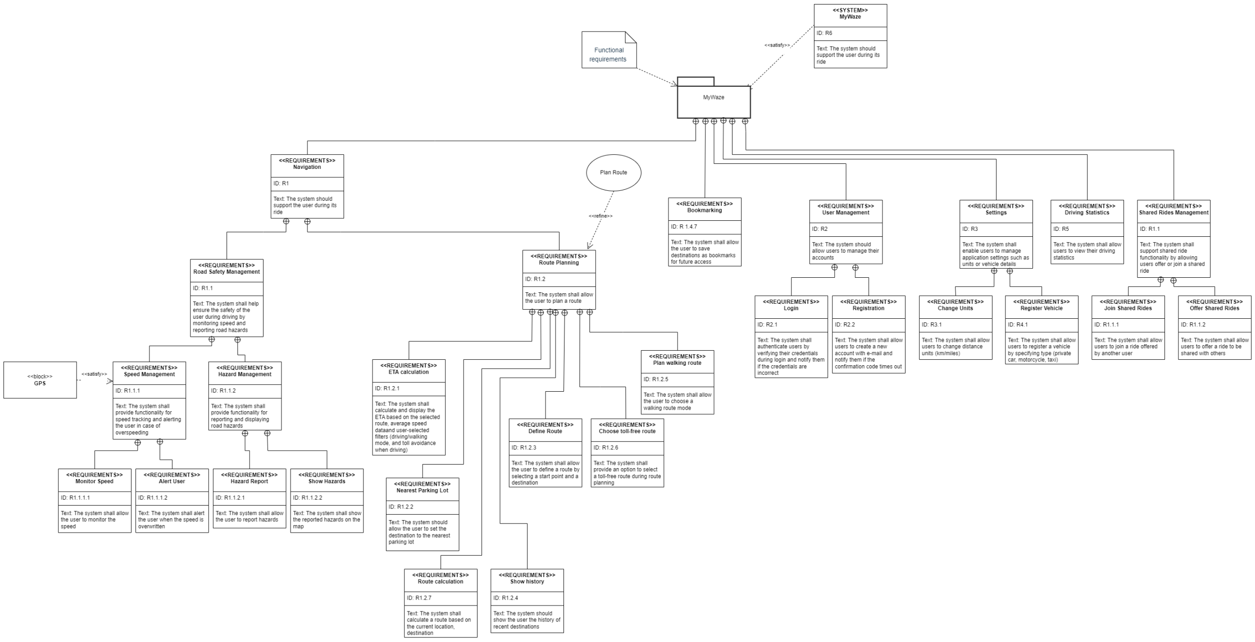


Figure 5: Functional Requirements

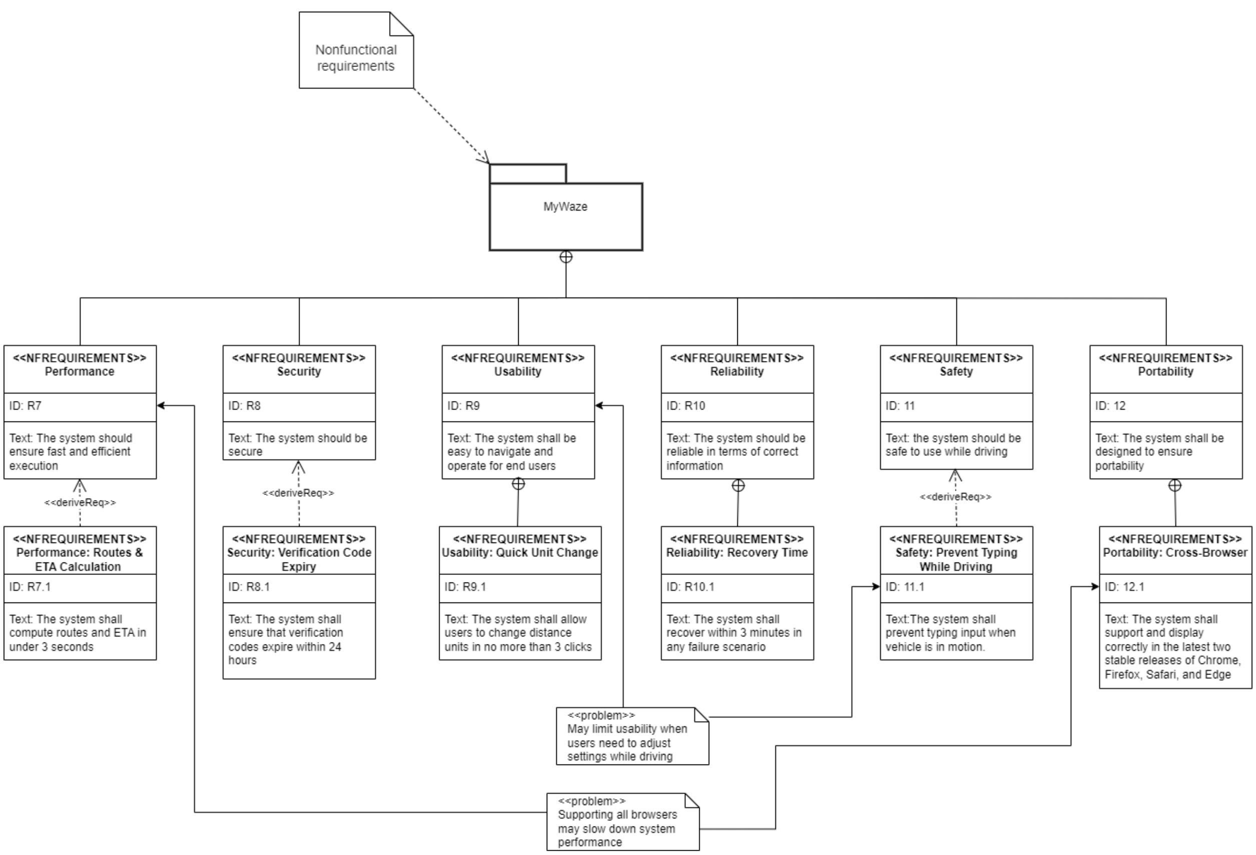


Figure 6: Nonfunctional Requirements

Mapping from Features to Models

Feature	Model
Overall process	MyWaze.bpmn
Register account	MyWaze_registerAccount.bpmn
Recent destinations	MyWaze_defineDestination.bpmn
Define route	MyWaze_defineRoute.bpmn, MyWaze_calculateRoute.bpmn
Avoid tolls	MyWaze_defineRoute.bpmn
Walking route	MyWaze_defineRoute.bpmn
ETA	MyWaze_defineRoute.bpmn
Current speed	MyWaze_navigate.bpmn, MyWaze_speedMonitoring.bpmn
Speed alert	MyWaze_navigate.bpmn, MyWaze_speedMonitoring.bpmn
Find parking lots	MyWaze_navigate.bpmn, MyWaze_defineRouteToParkingLot.bpmn, MyWaze_calculateRoute.bpmn
Hazard reports	MyWaze_navigate.bpmn, MyWaze_reportHazard.bpmn
Share ride	MyWaze.bpmn, MyWaze_registerRideOffering.bpmn, MyWaze_arrangeSharedRide.bpmn
Bookmarking	MyWaze.bpmn, MyWaze_createBookmark.bpmn
Statistics	MyWaze.bpmn, MyWaze_provideStatistics.bpmn
Change units	MyWaze.bpmn, MyWaze_changeUnits.bpmn
Register vehicle type	MyWaze.bpmn, MyWaze_registerVehicleType.bpmn

BPMN Models

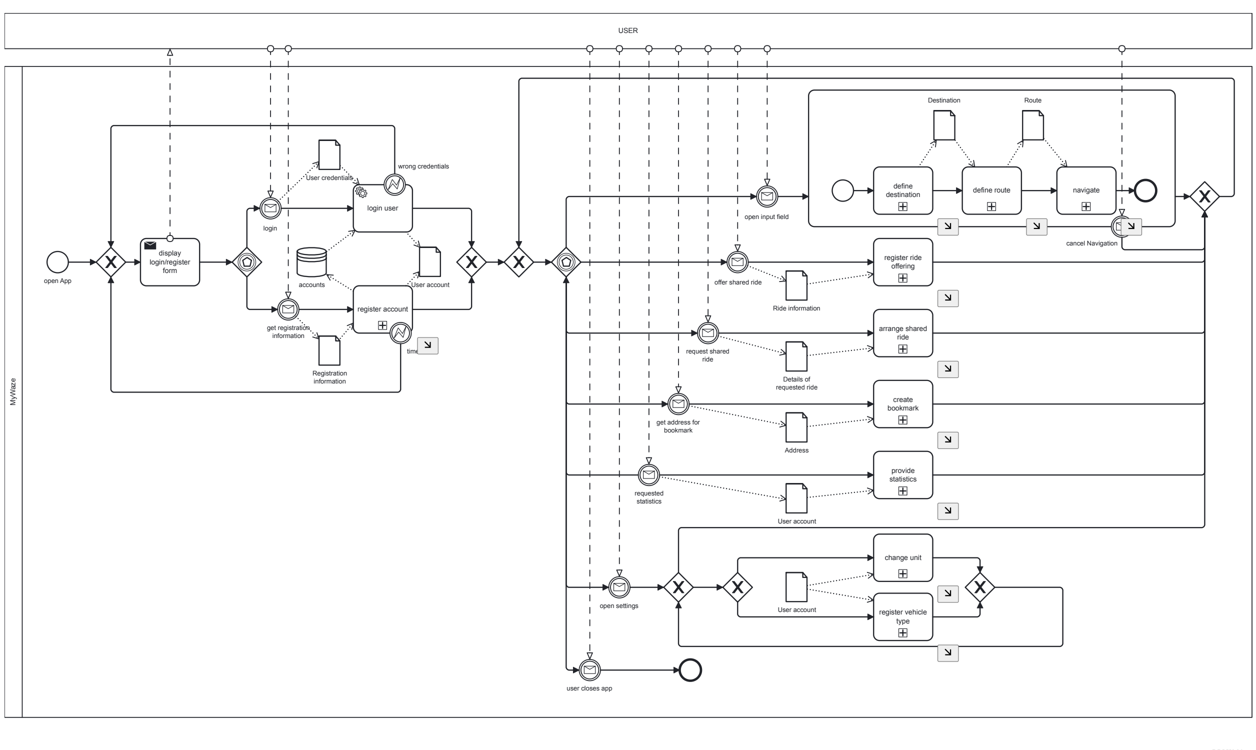


Figure 7: Model: MyWaze

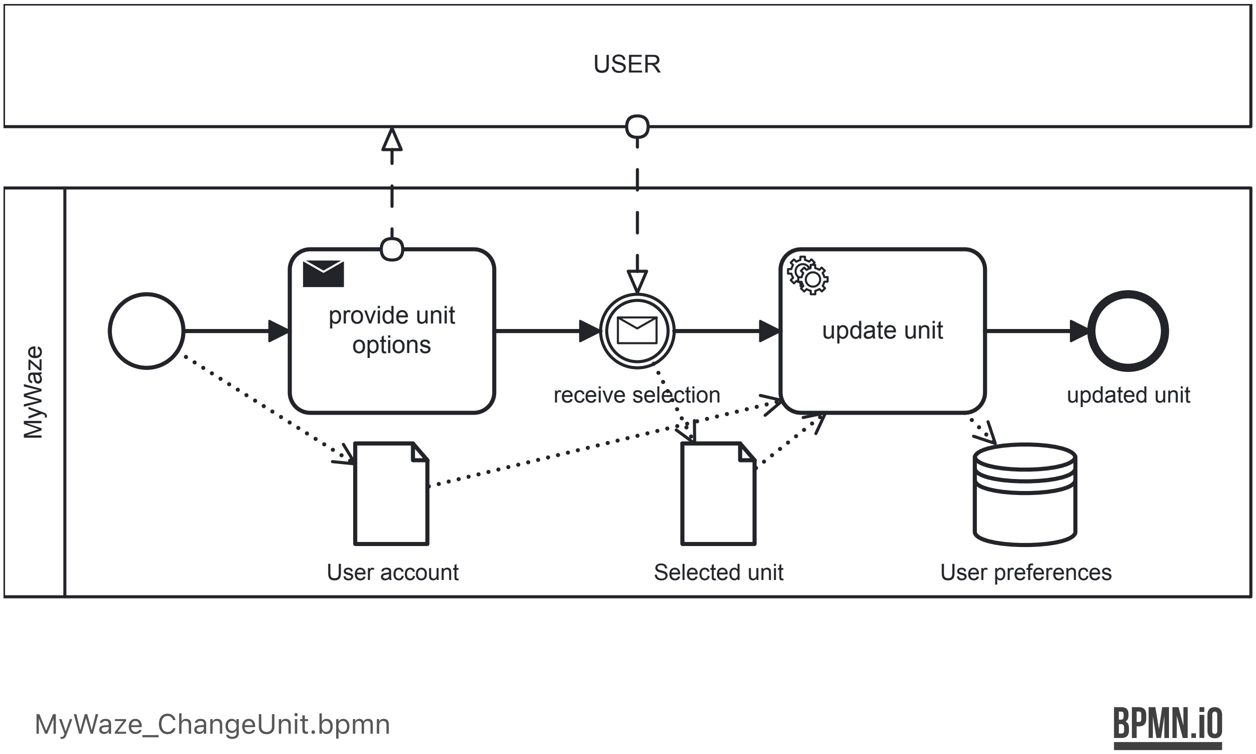


Figure 8: Model: MyWaze˙ChangeUnit

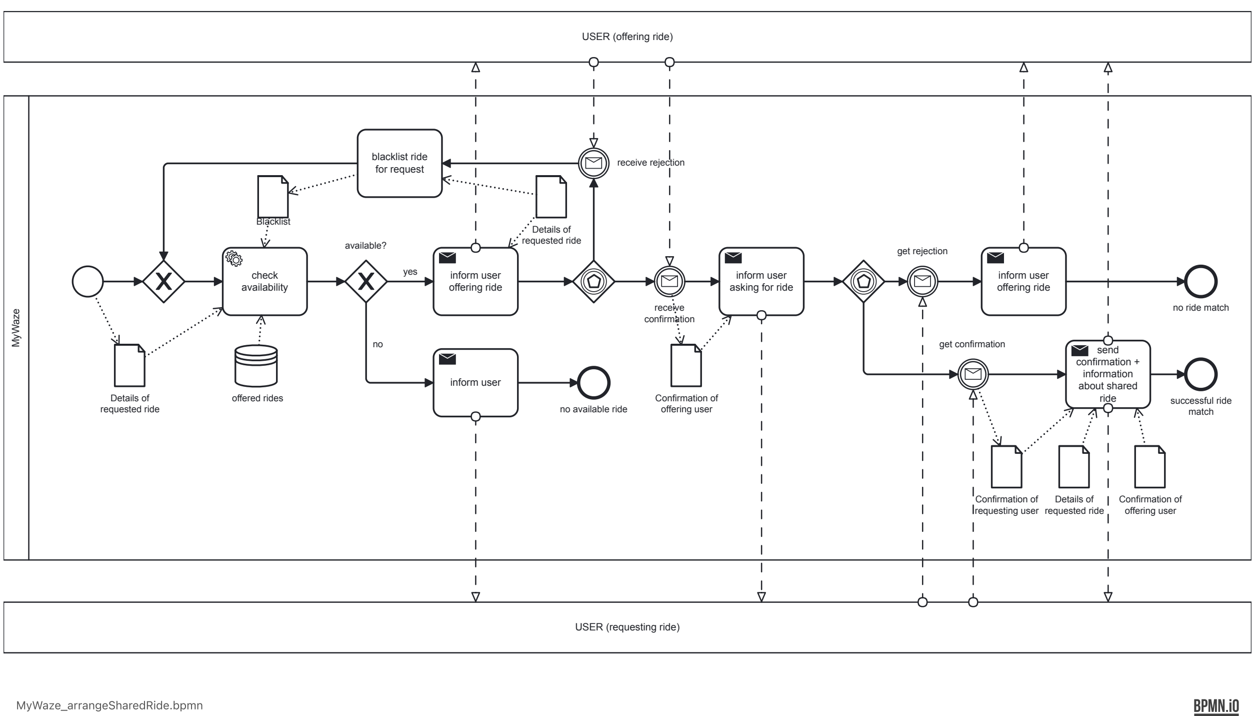


Figure 9: Model: MyWaze˙arrangeSharedRide

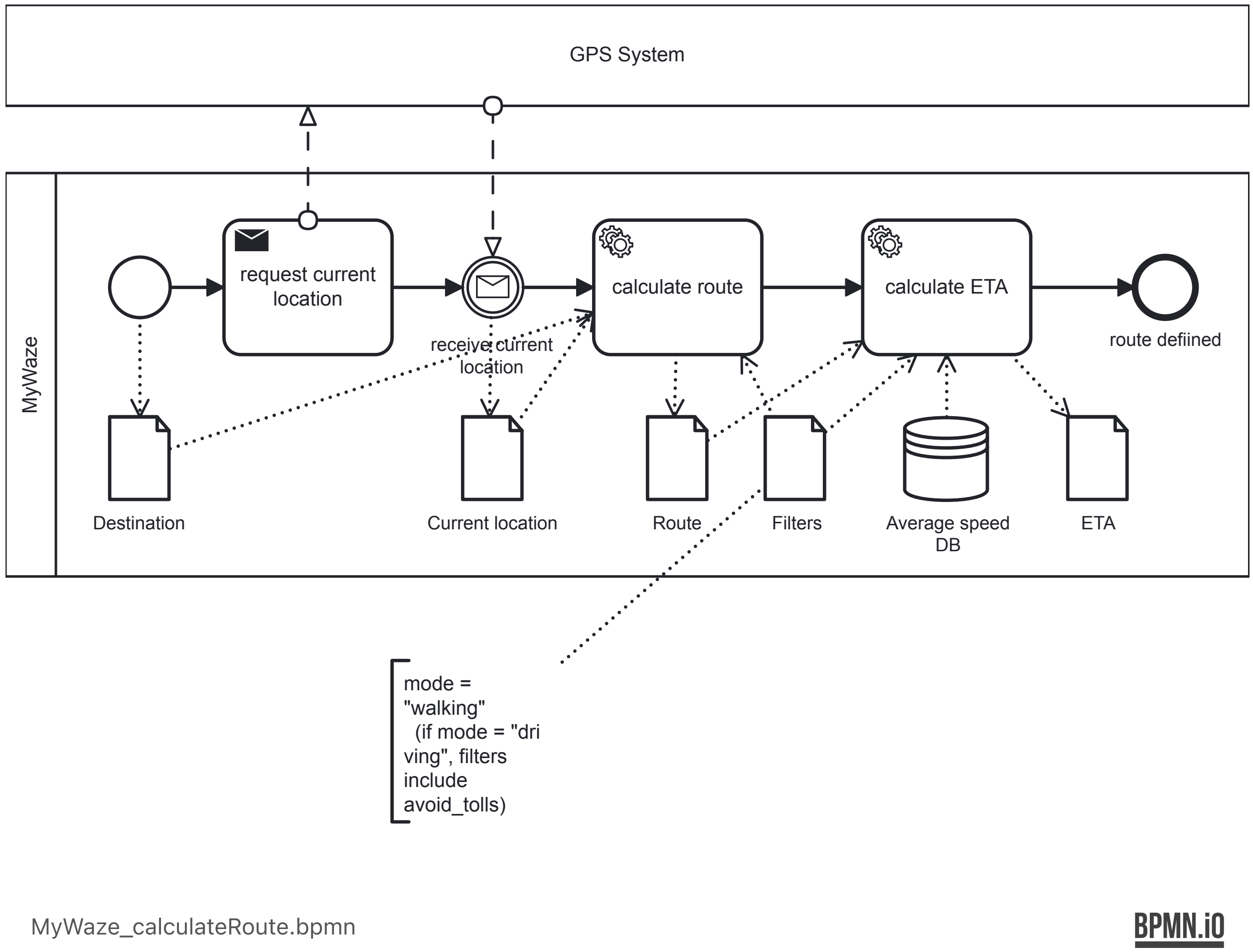


Figure 10: Model: MyWaze˙calculateRoute

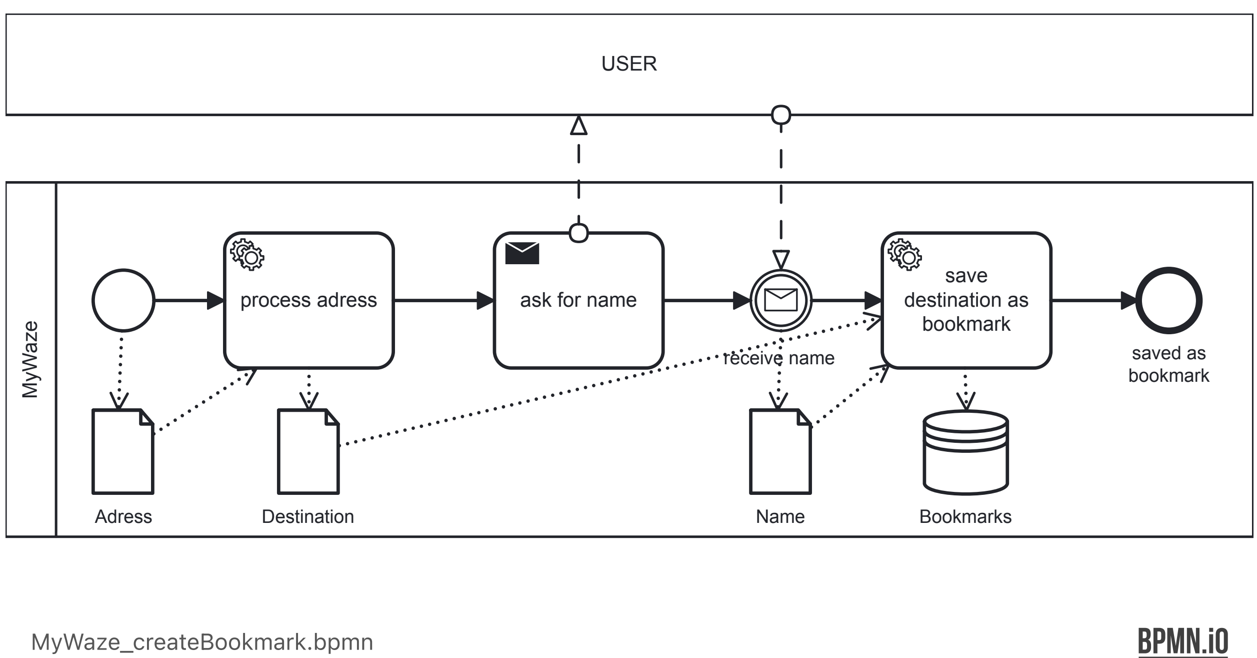


Figure 11: Model: MyWaze˙createBookmark

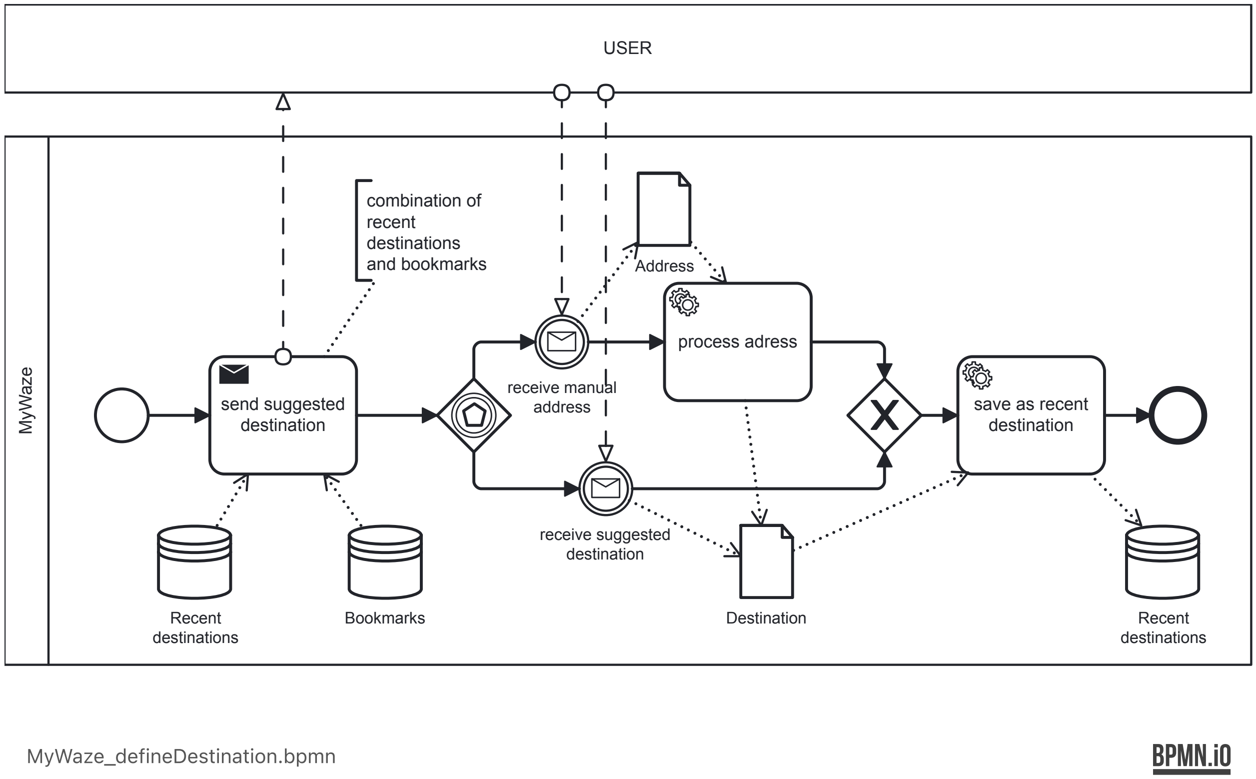


Figure 12: Model: MyWaze˙defineDestination

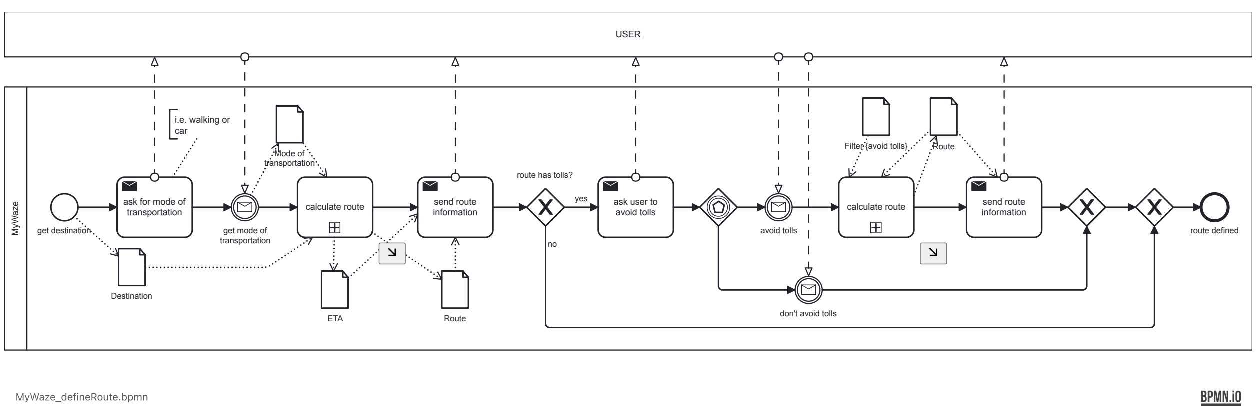


Figure 13: Model: MyWaze˙defineRoute

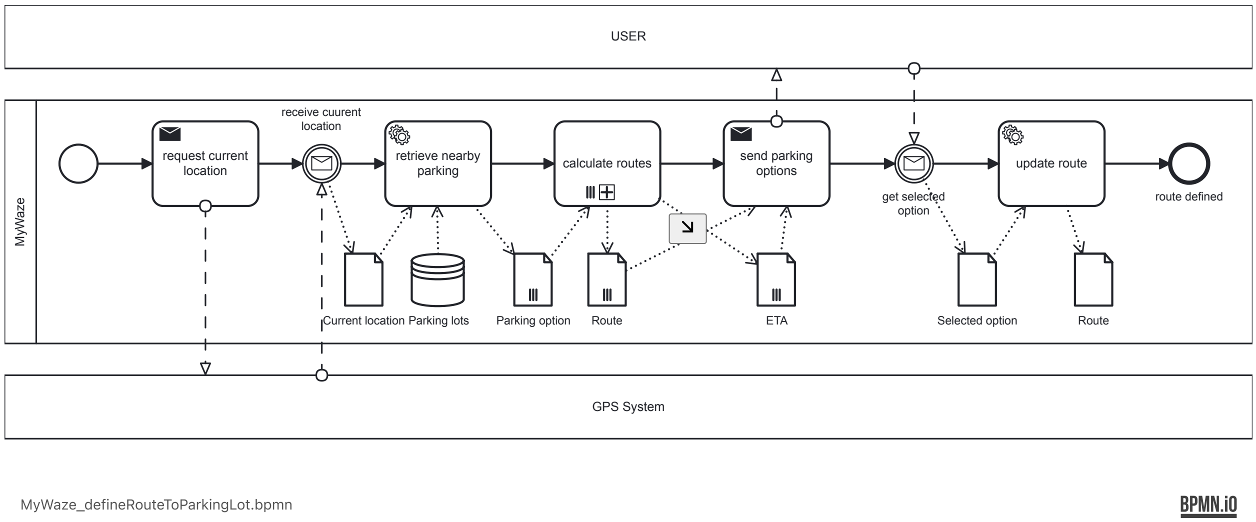


Figure 14: Model: MyWaze˙defineRouteToParkingLot

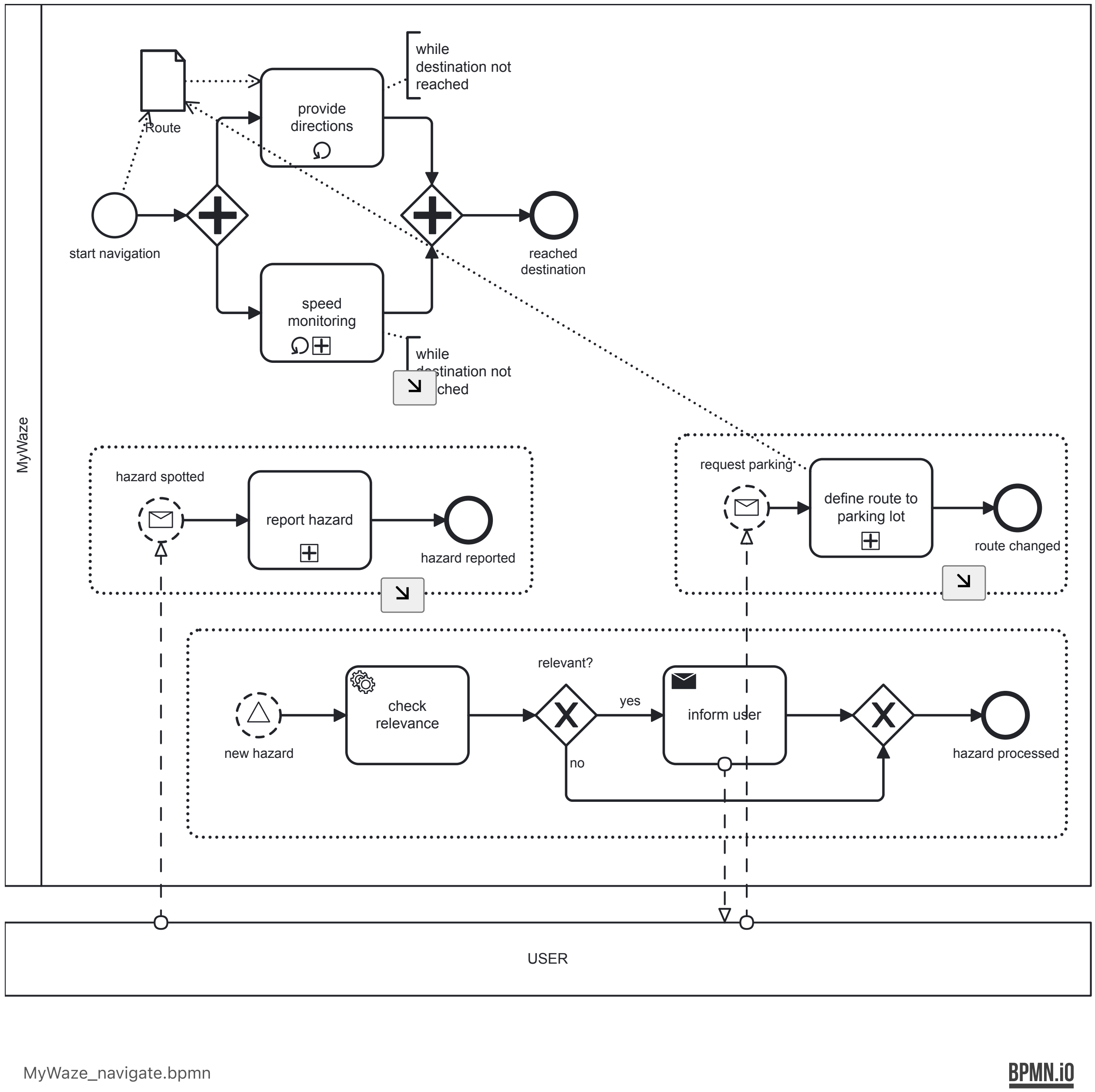


Figure 15: Model: MyWaze˙navigate

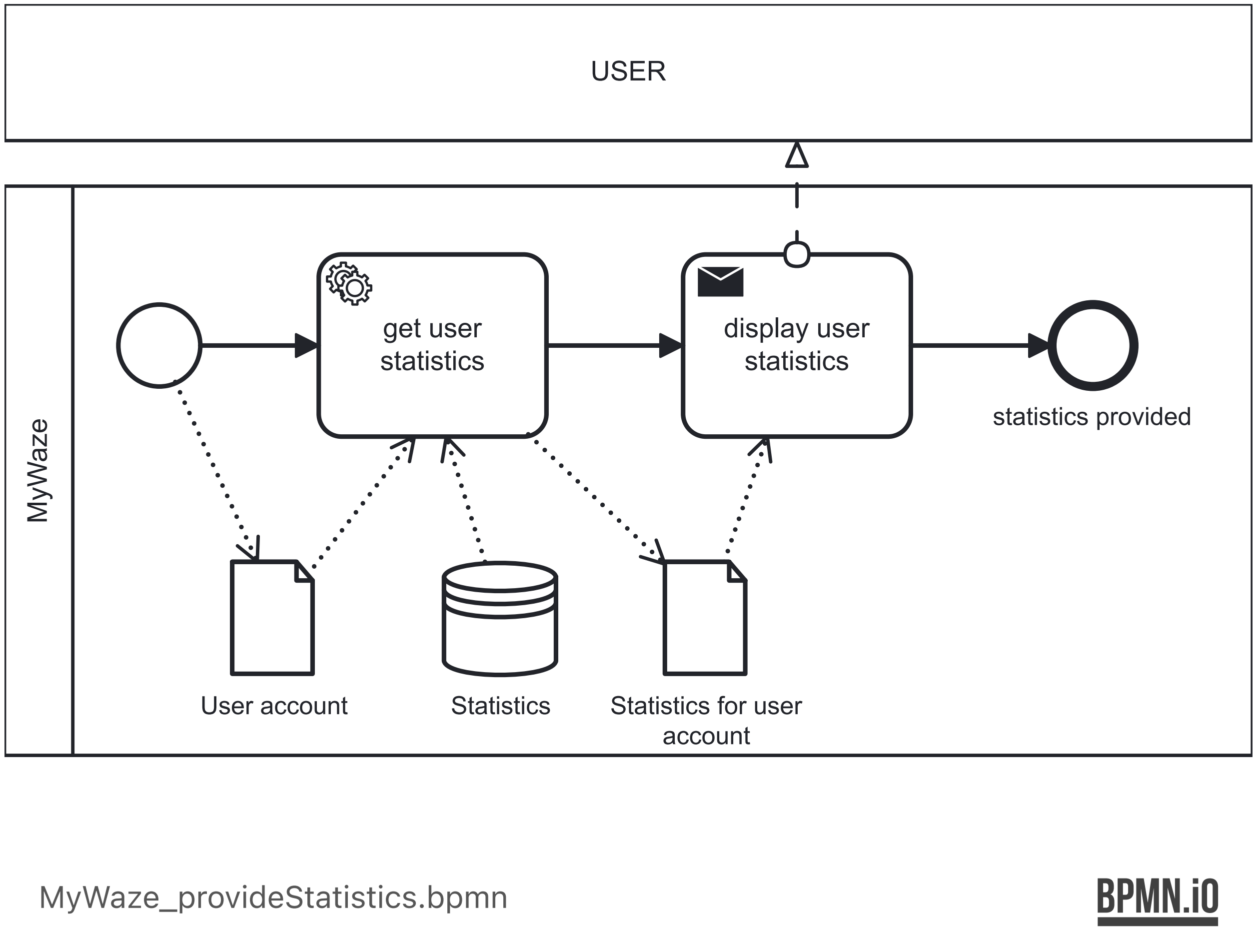


Figure 16: Model: MyWaze˙provideStatistics

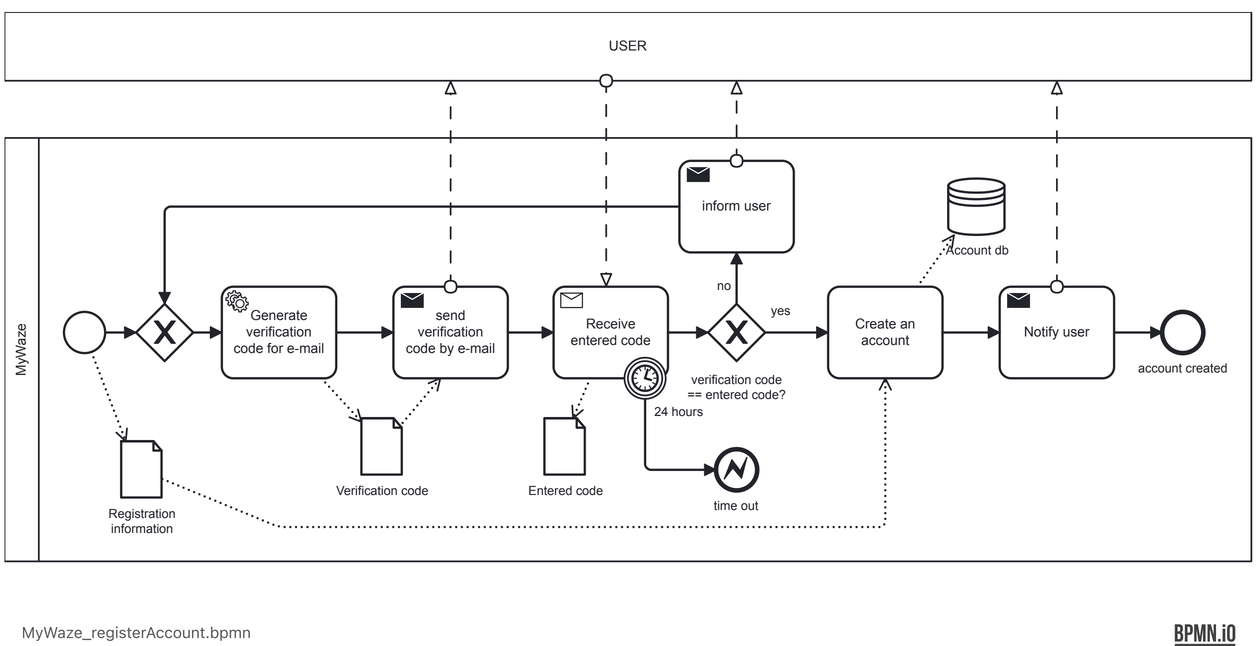


Figure 17: Model: MyWaze˙registerAccount

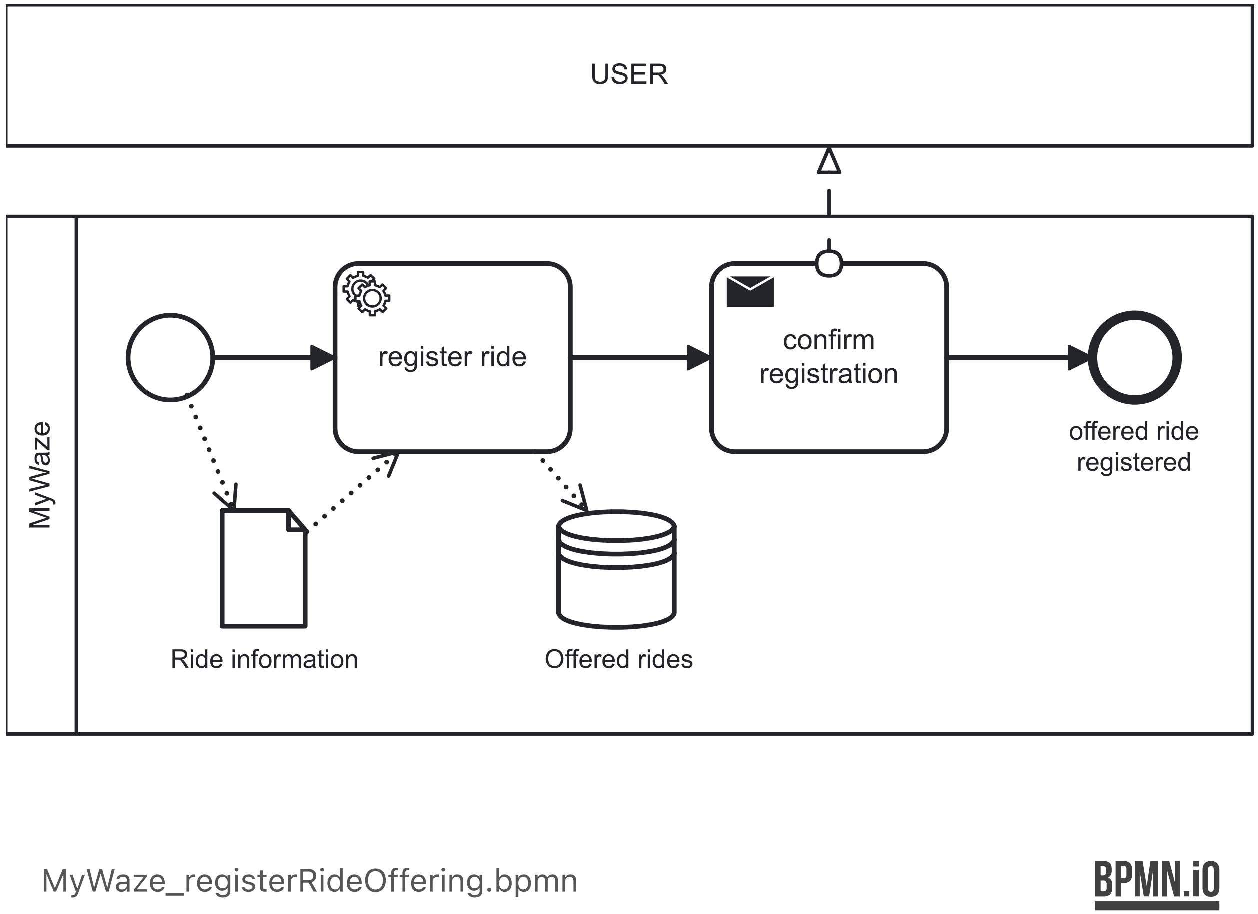


Figure 18: Model: MyWaze˙registerRideOffering

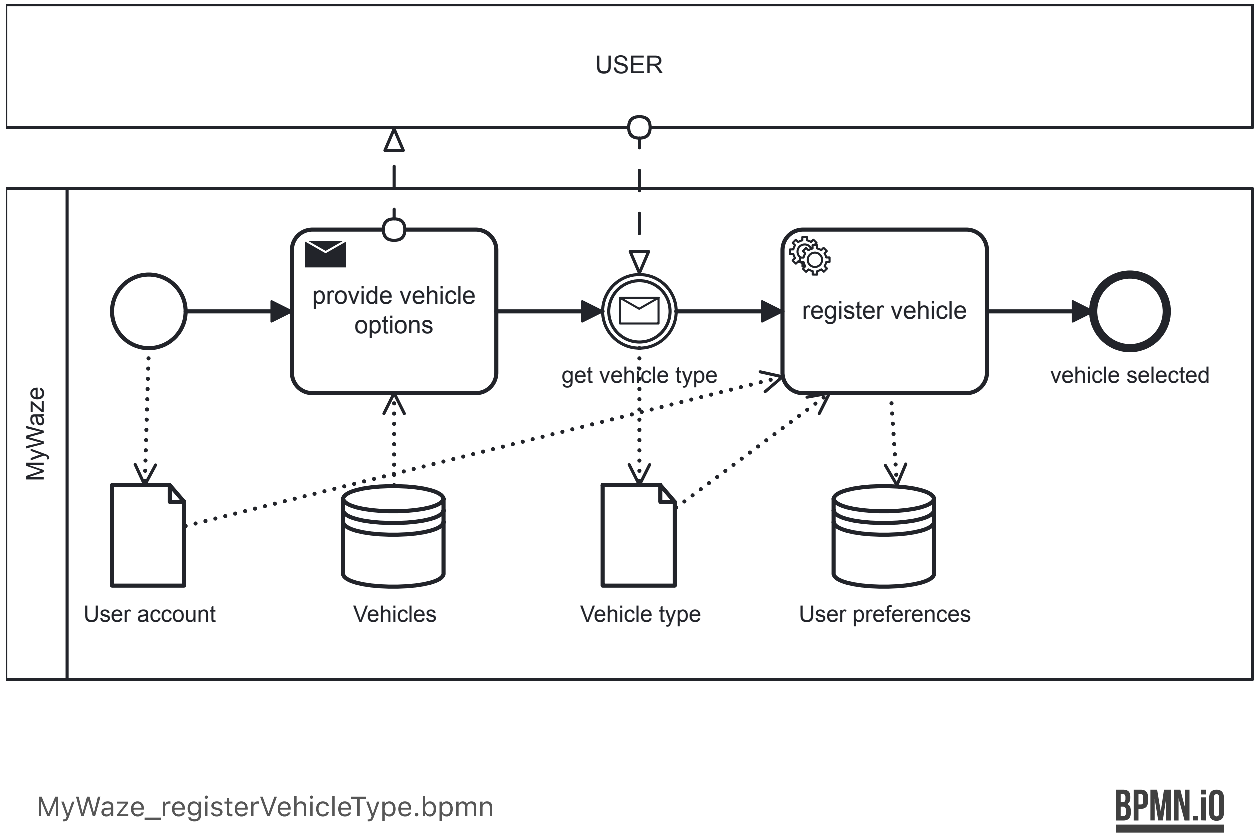


Figure 19: Model: MyWaze˙registerVehicleType

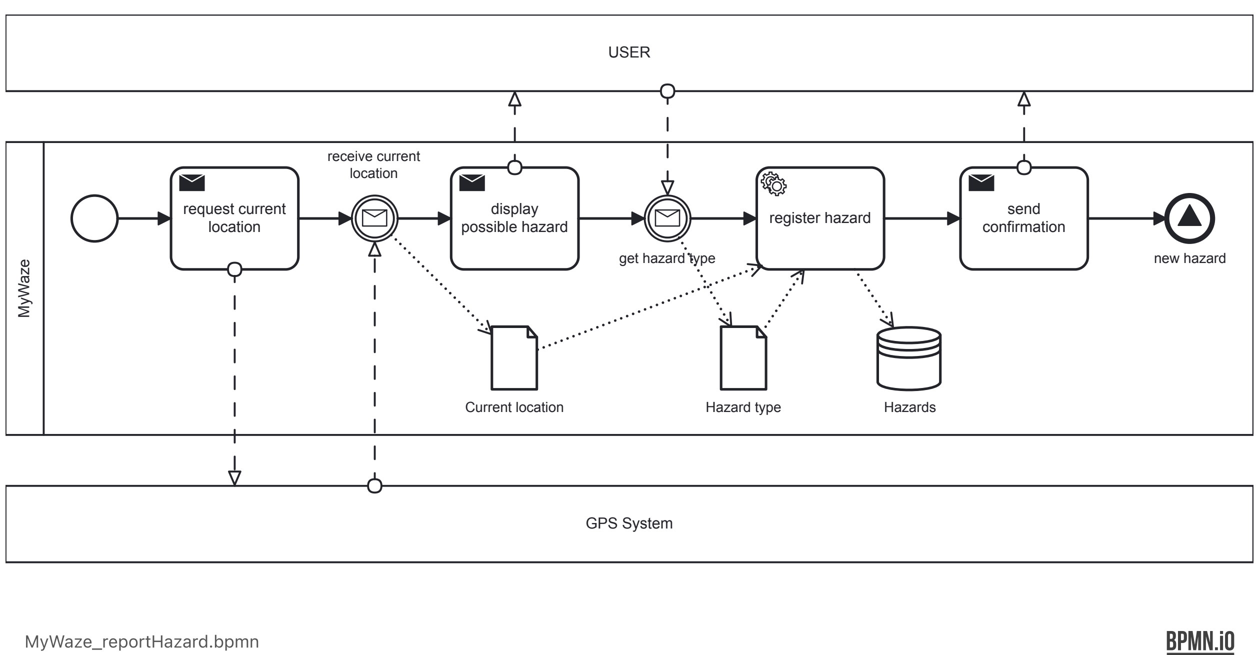


Figure 20: Model: MyWaze˙reportHazard

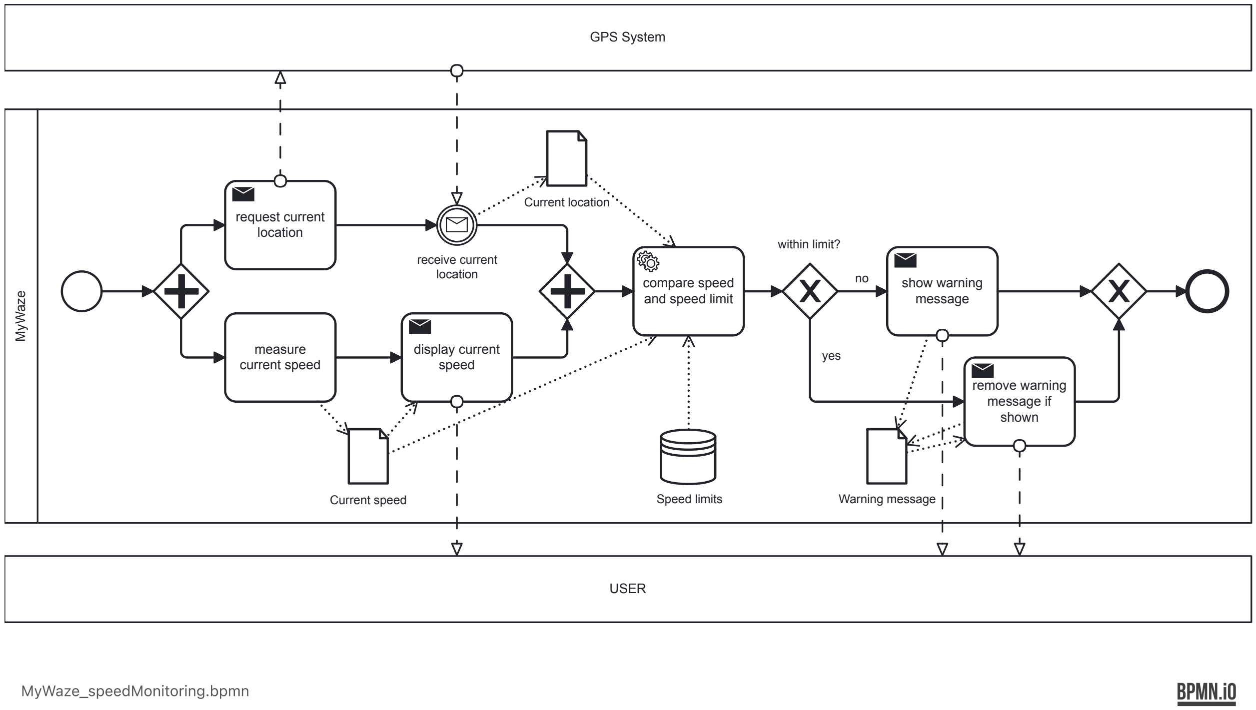


Figure 21: Model: MyWaze˙speedMonitoring

Context Diagram

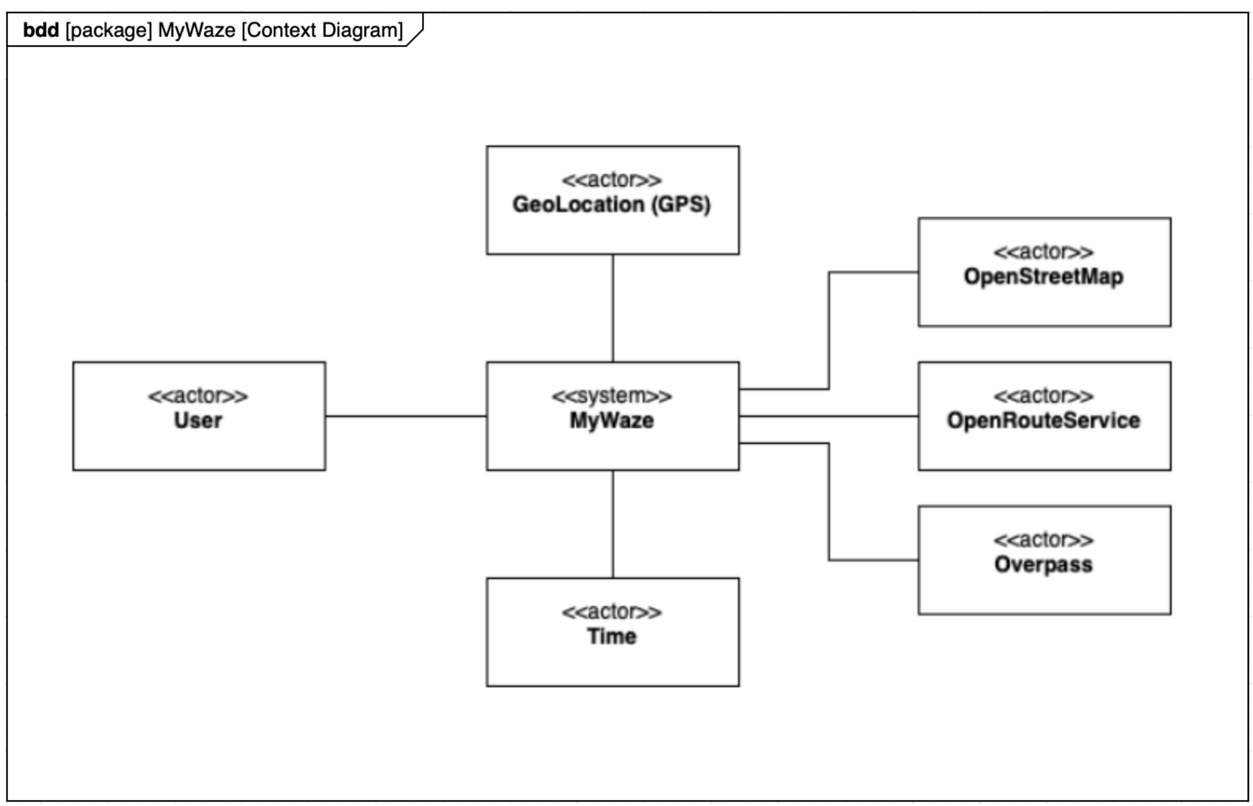


Figure 22: Context Diagram

Use Case View

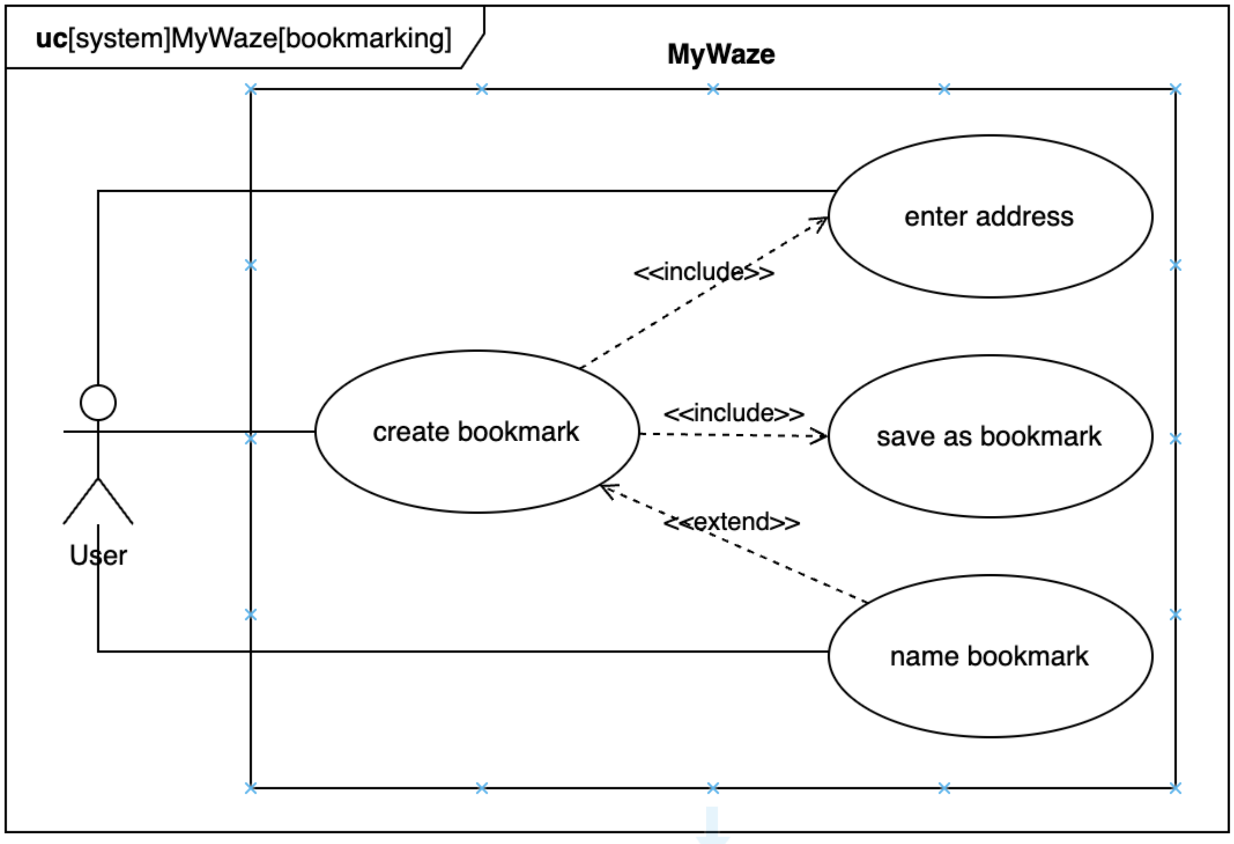


Figure 23: Use Case Diagram: Bookmarking

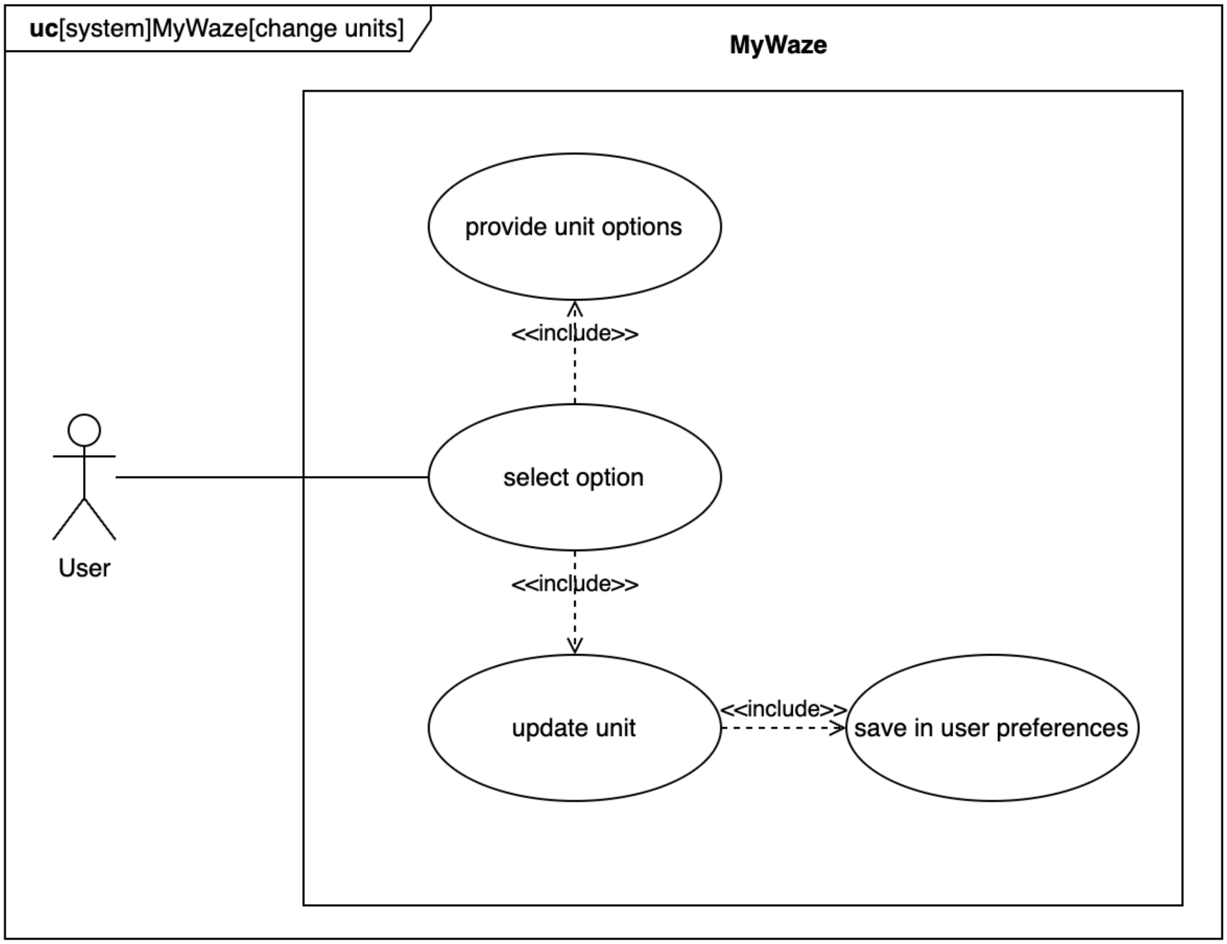


Figure 24: Use Case Diagram: Change Units

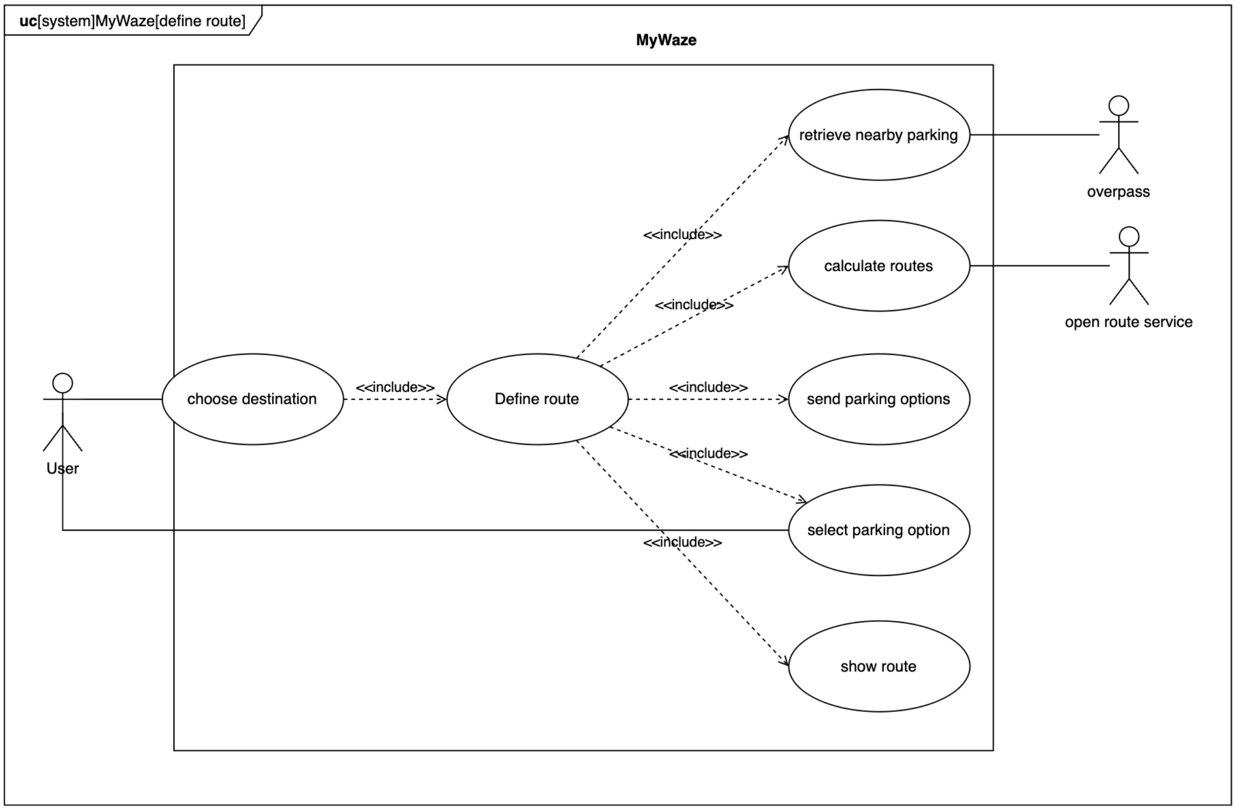


Figure 25: Use Case Diagram: Define Route

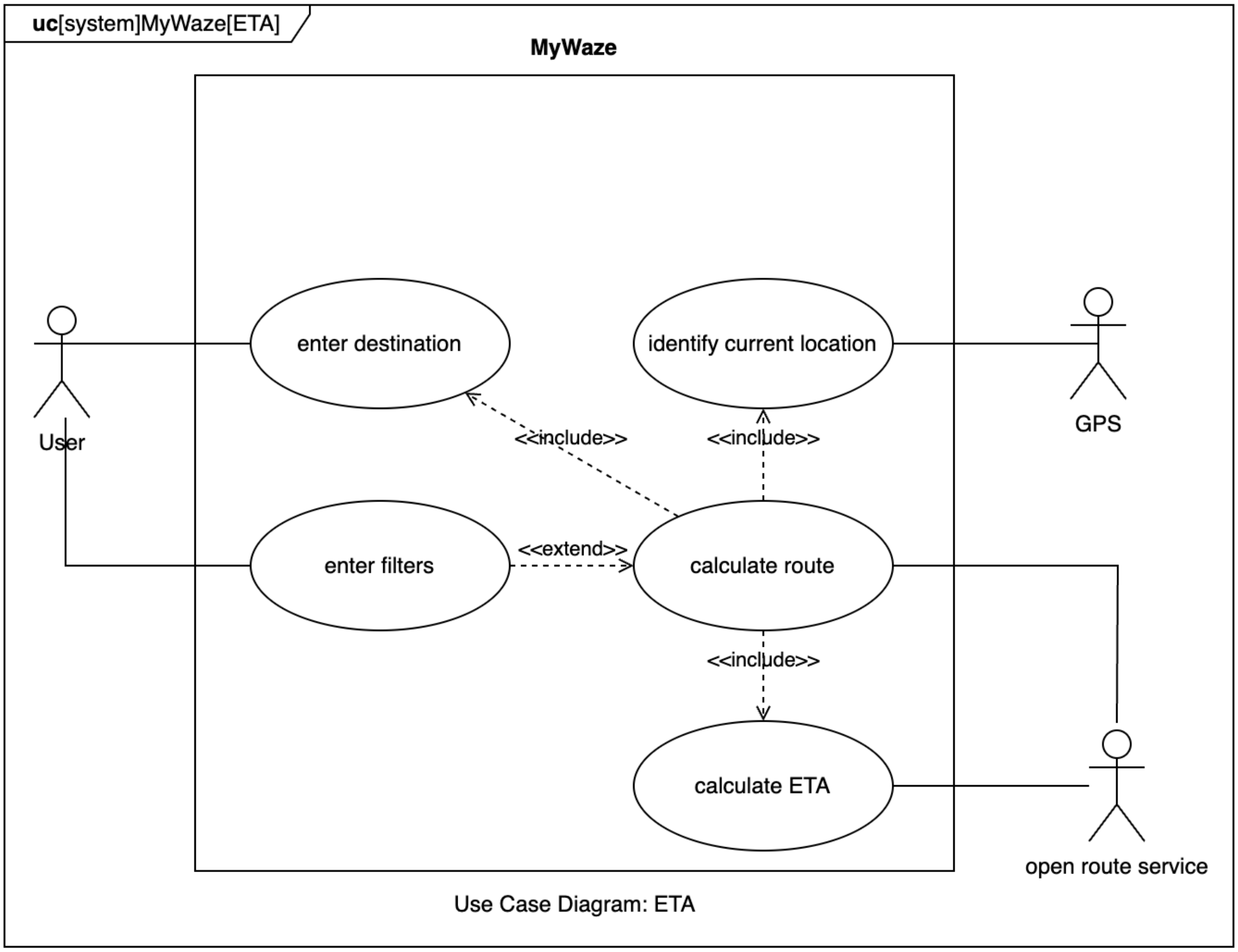


Figure 26: Use Case Diagram: ETA

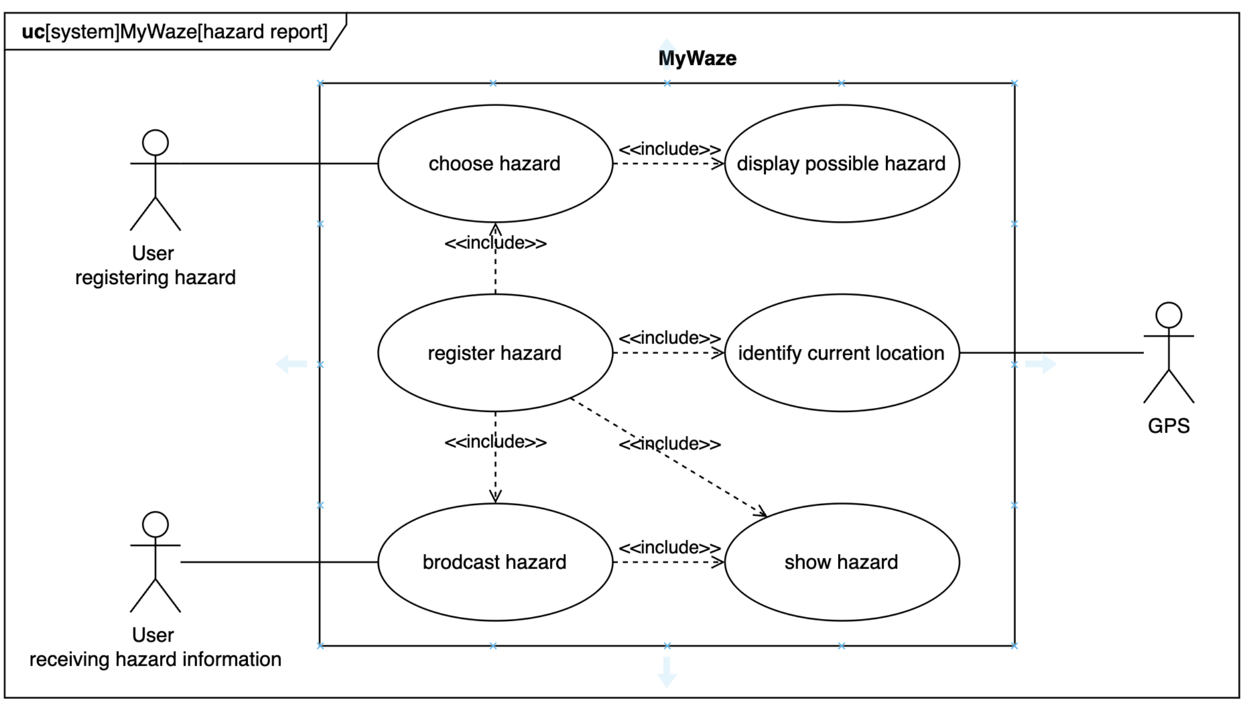


Figure 27: Use Case Diagram: Hazard Report

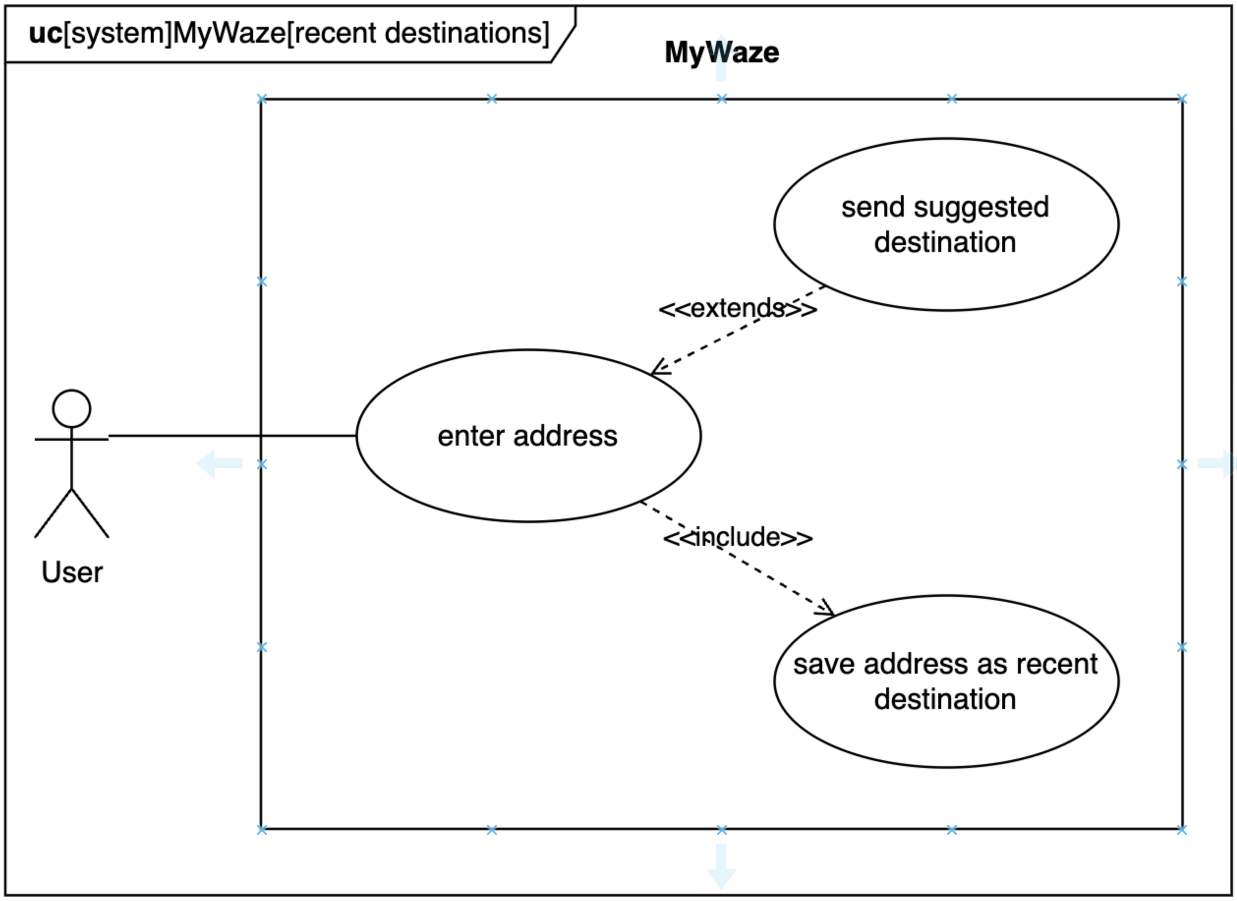


Figure 28: Use Case Diagram: Recent Destinations

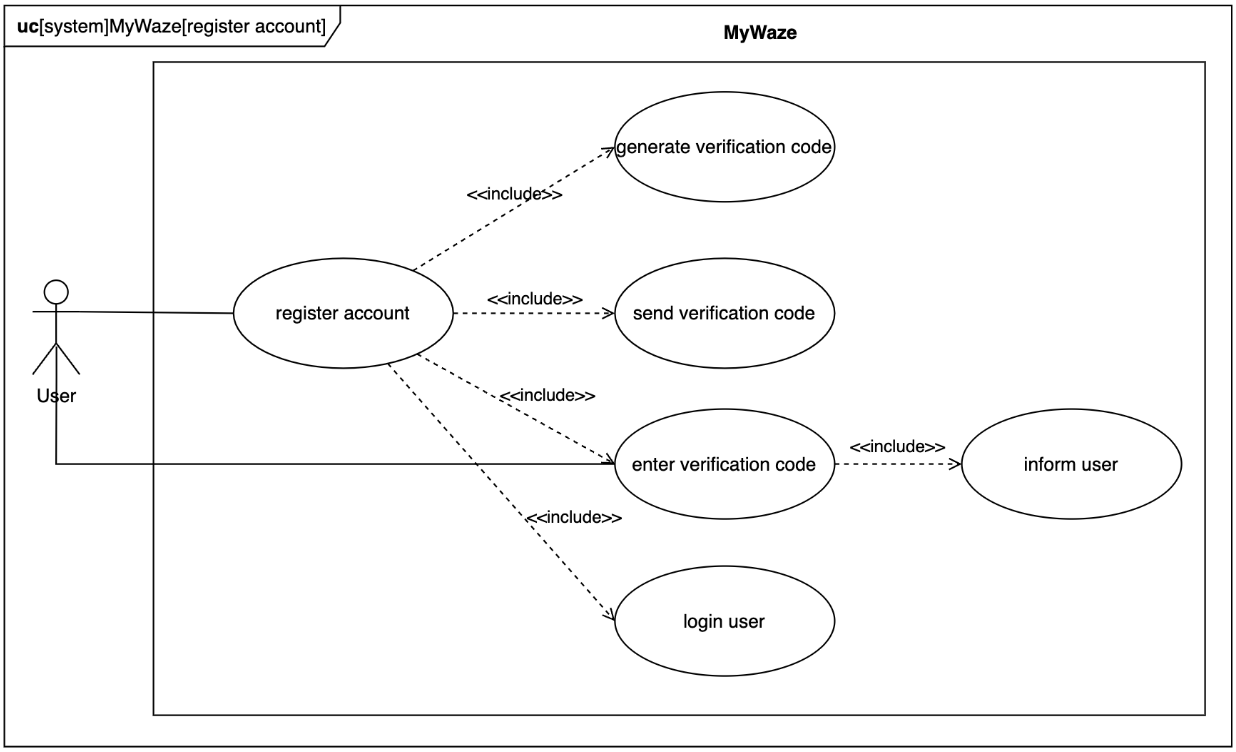


Figure 29: Use Case Diagram: Register Account

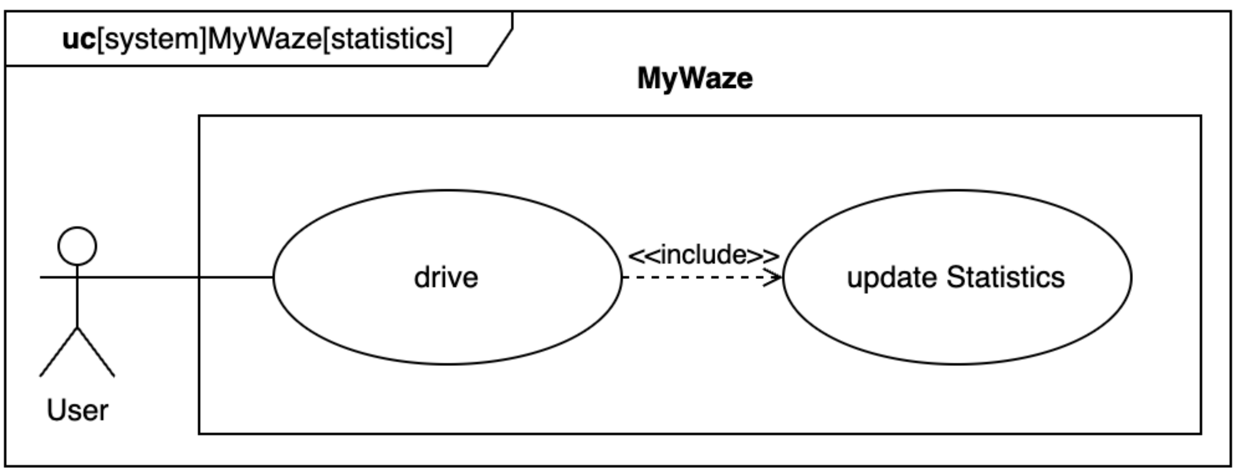


Figure 30: Use Case Diagram: Statistics

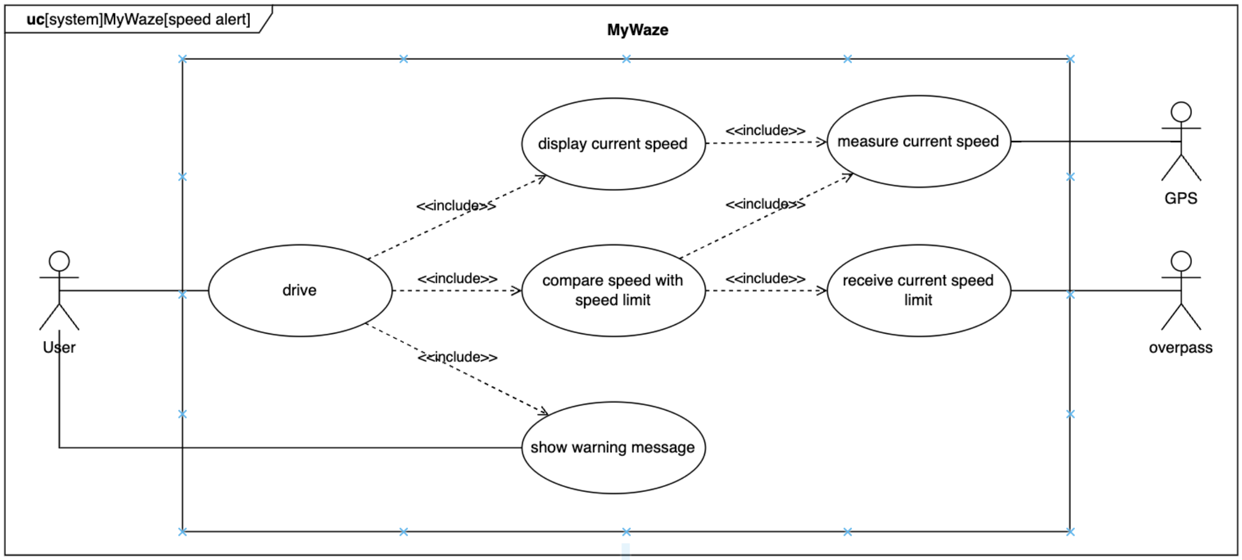


Figure 31: Use Case Diagram: Speed Alert

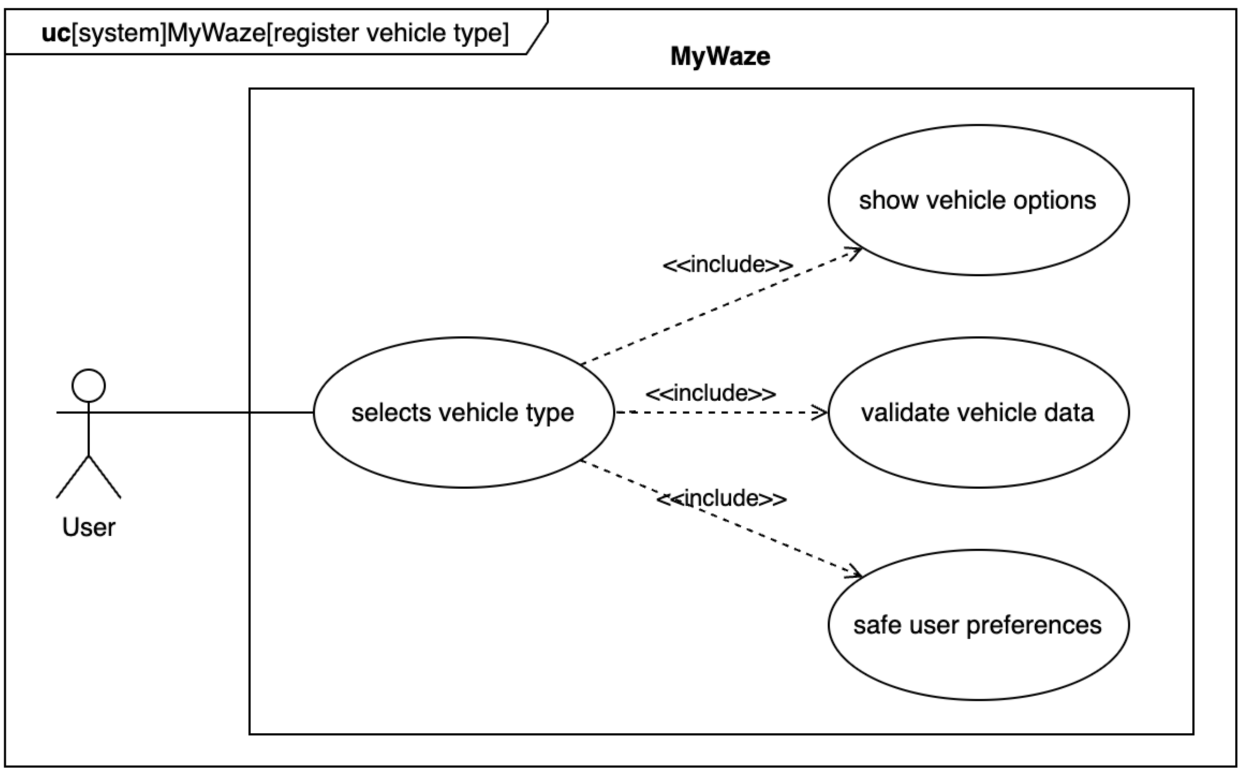


Figure 32: Use Case Diagram: Register Vehicle Type

Logical View

System Decomposition

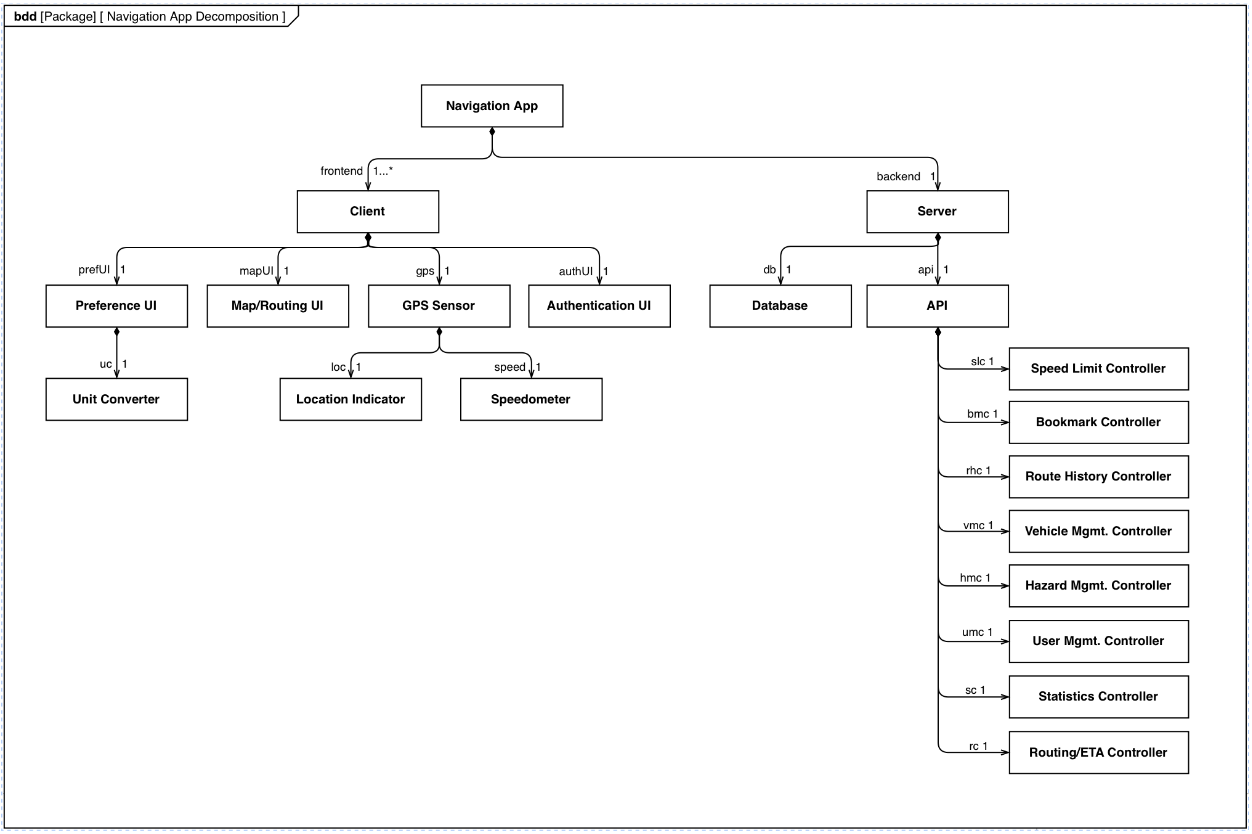


Figure 33: MyWaze Decomposition

Navigation Domain

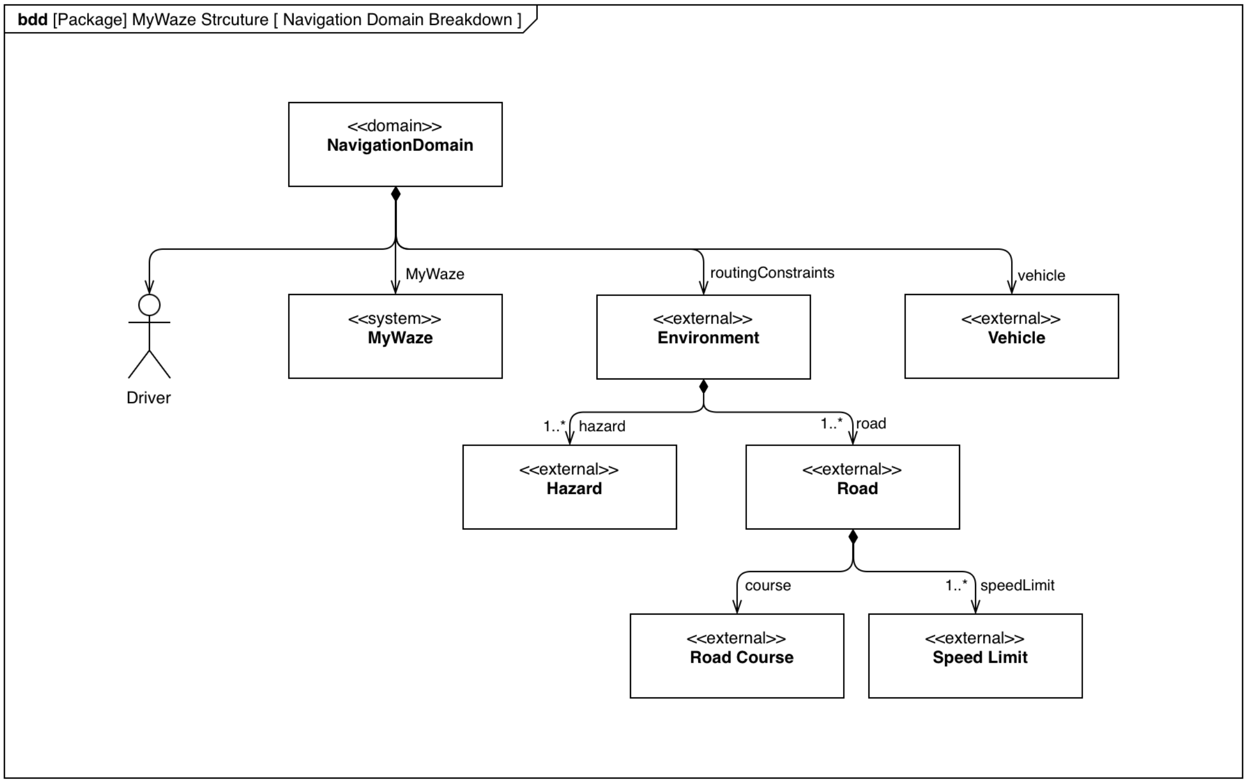


Figure 34: Navigation Domain

Process View

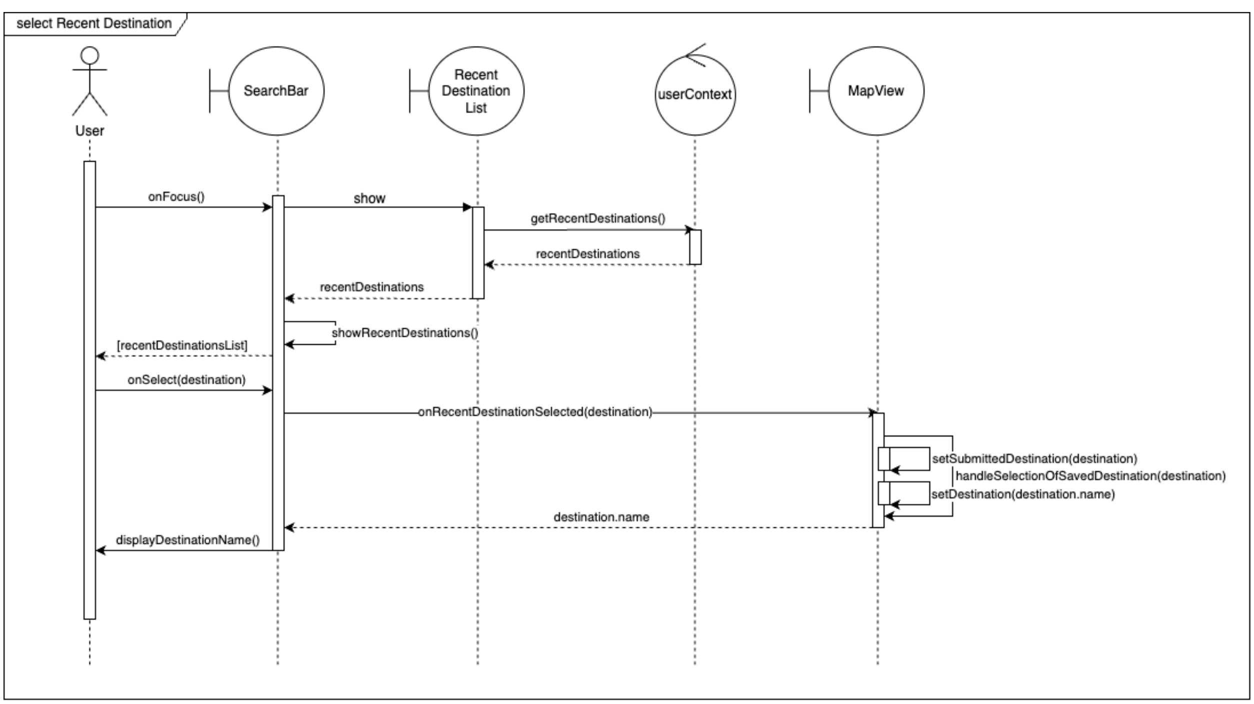


Figure 35: Sequence Diagram: Recent Destinations

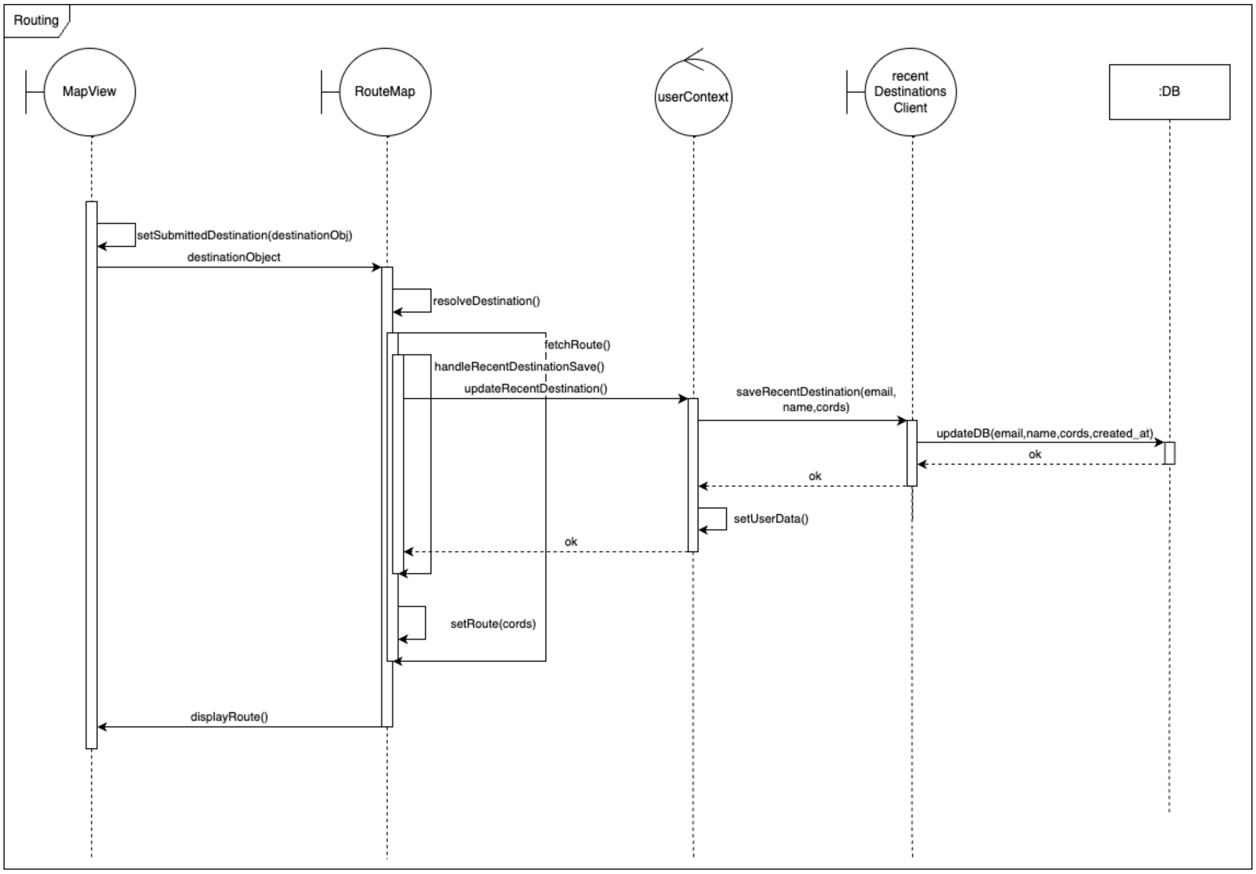


Figure 36: Sequence Diagram: Routing

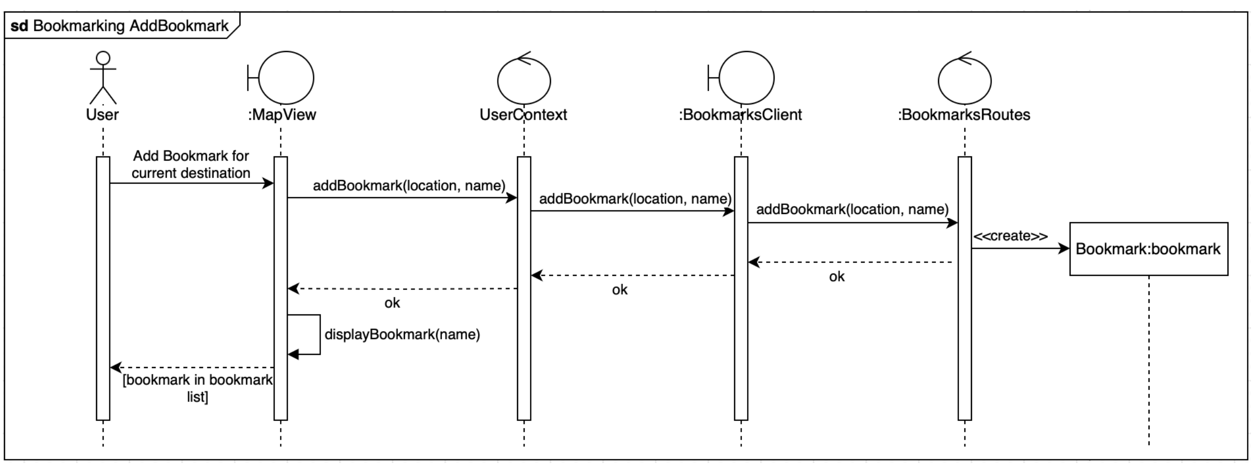


Figure 37: Sequence Diagram: Bookmarking (add)

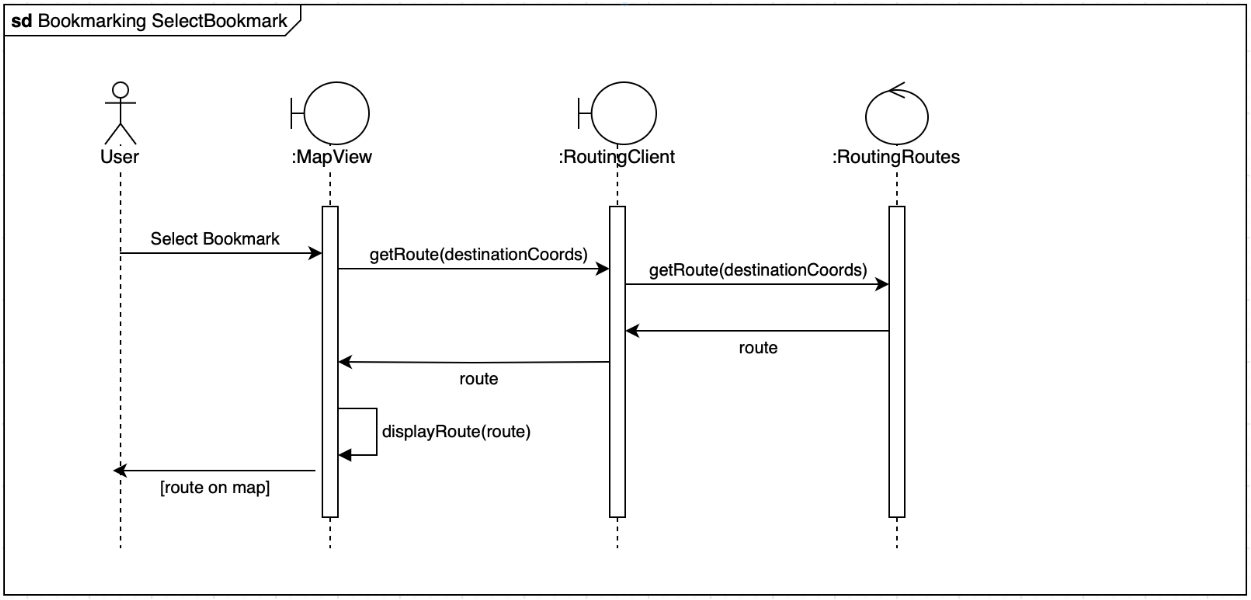


Figure 38: Sequence Diagram: Bookmarking (select)

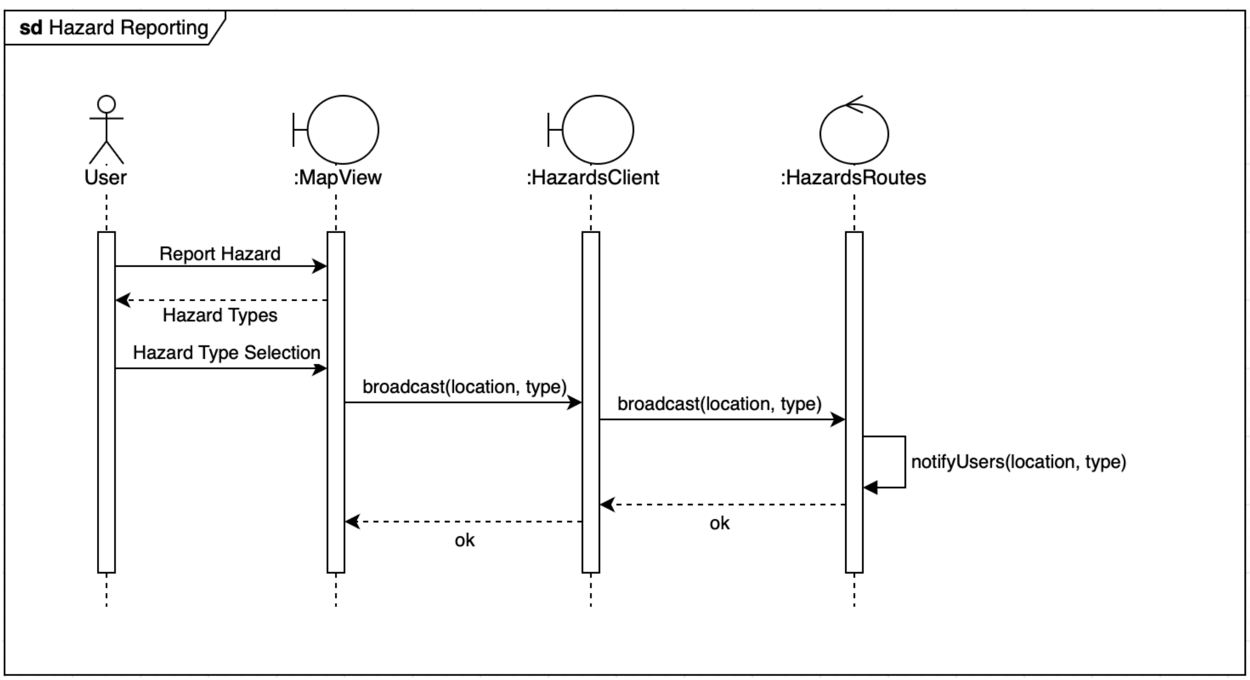


Figure 39: Sequence Diagram: Hazards (report)

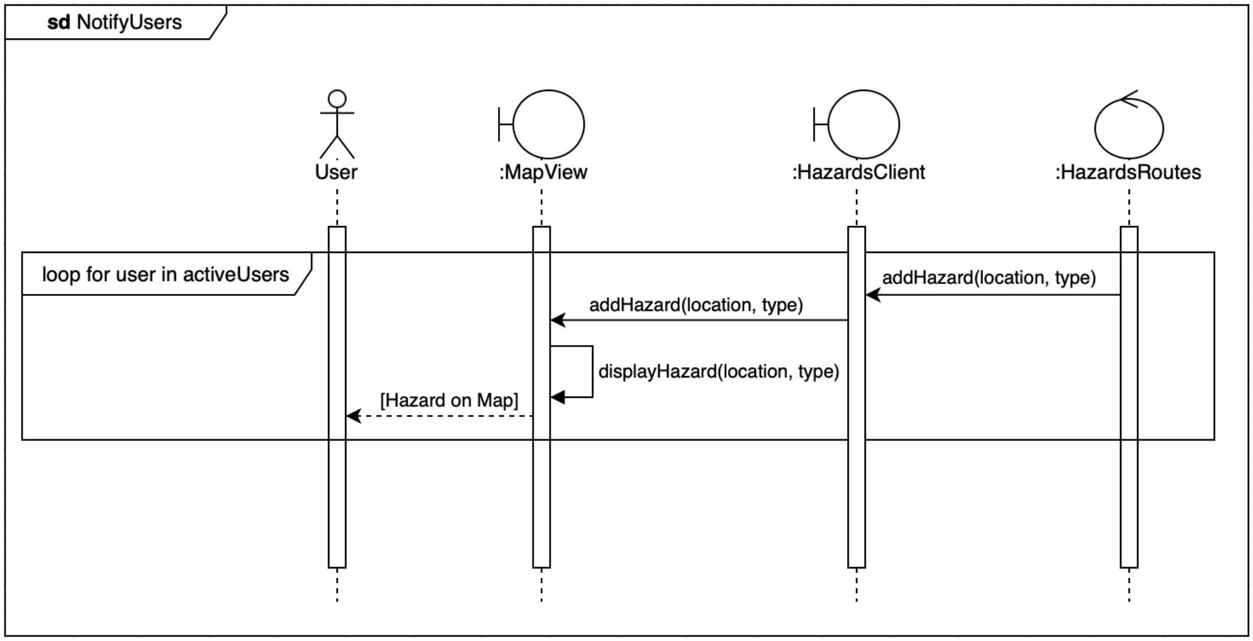


Figure 40: Sequence Diagram: Hazards (notify)

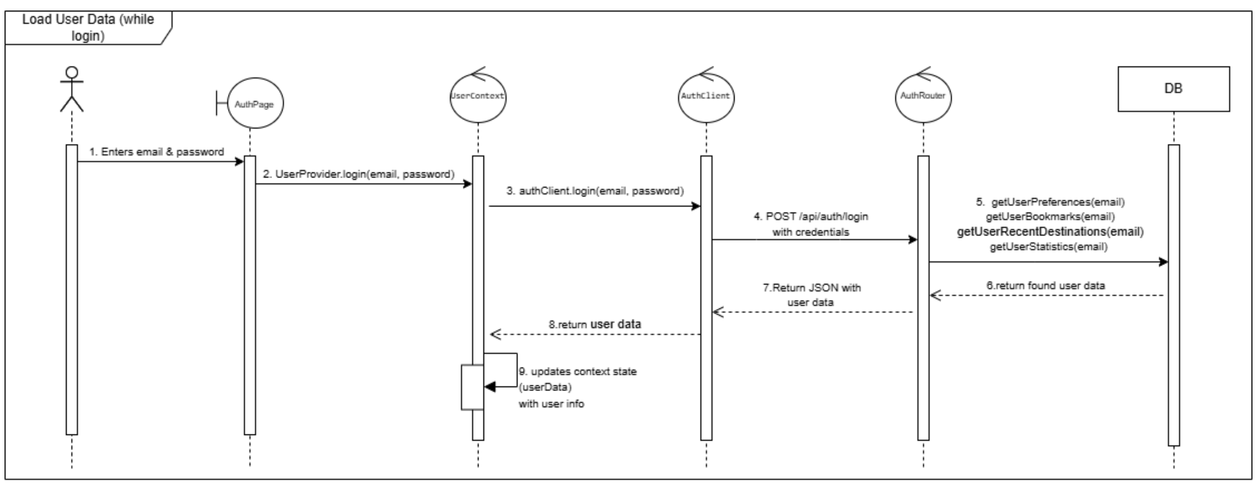


Figure 41: Sequence Diagram: User Data

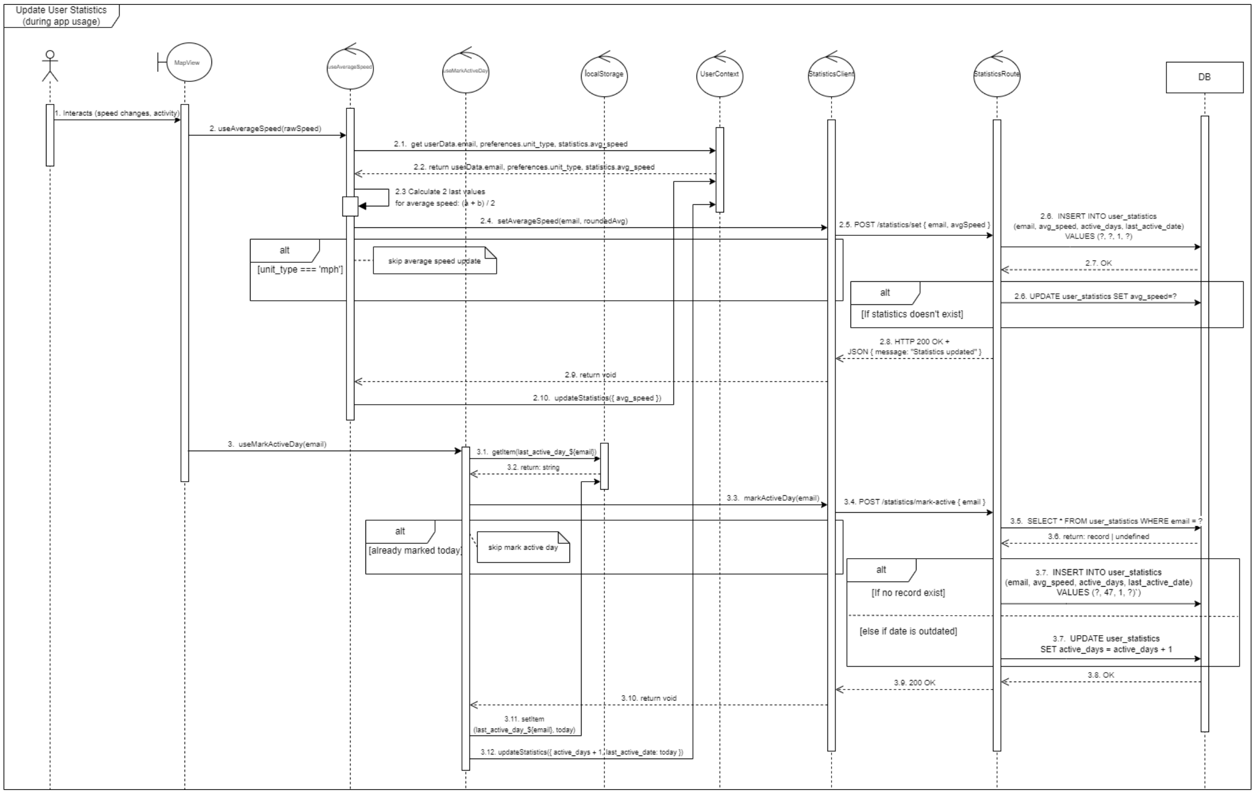


Figure 42: Sequence Diagram: Update Statistics

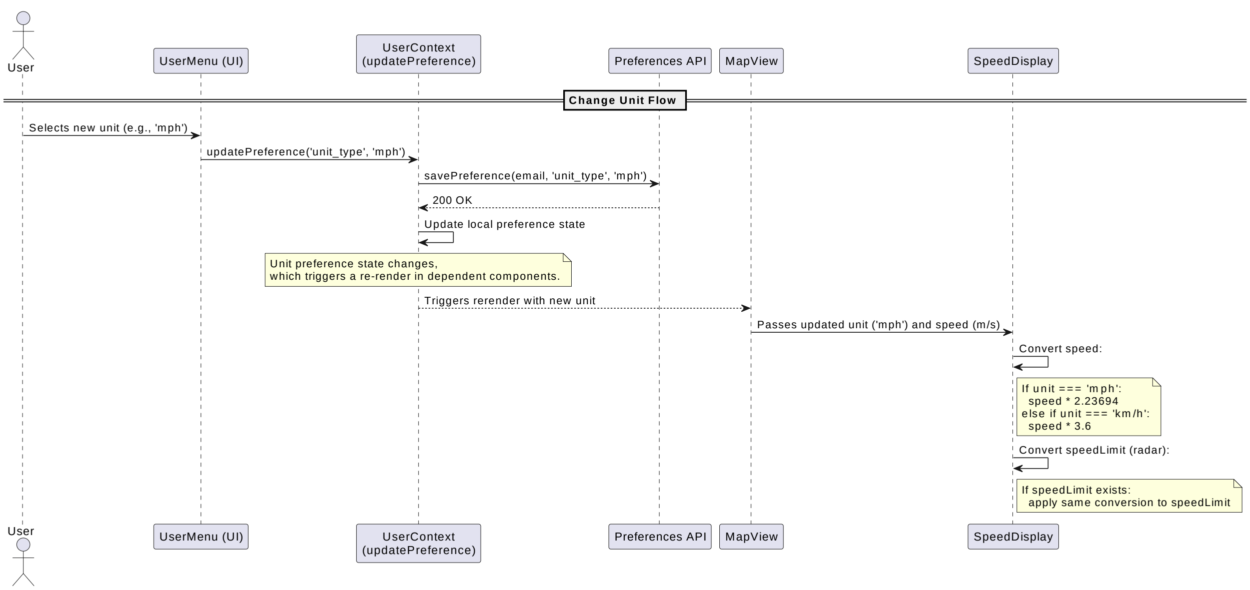


Figure 43: Sequence Diagram: Change Units

Development View

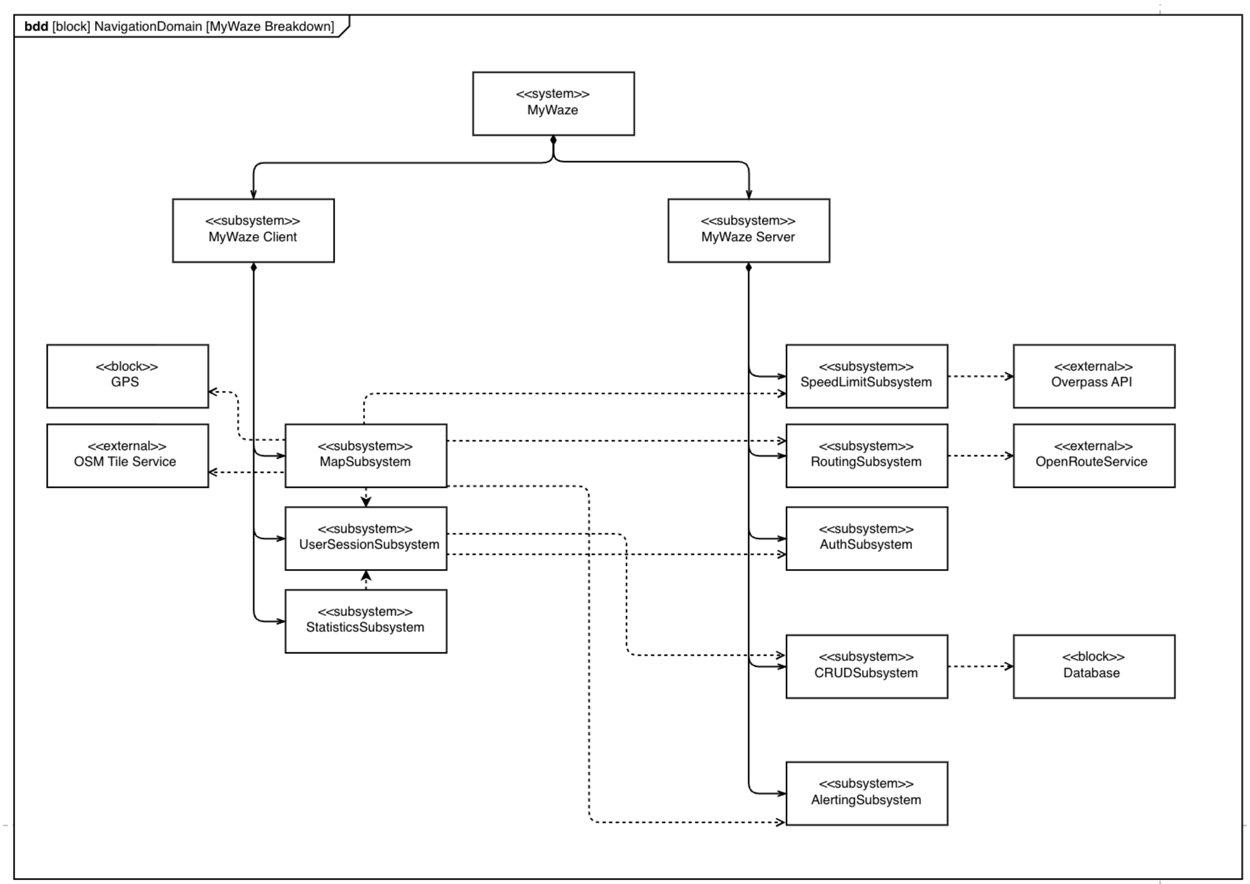


Figure 44: MyWaze Subsystems

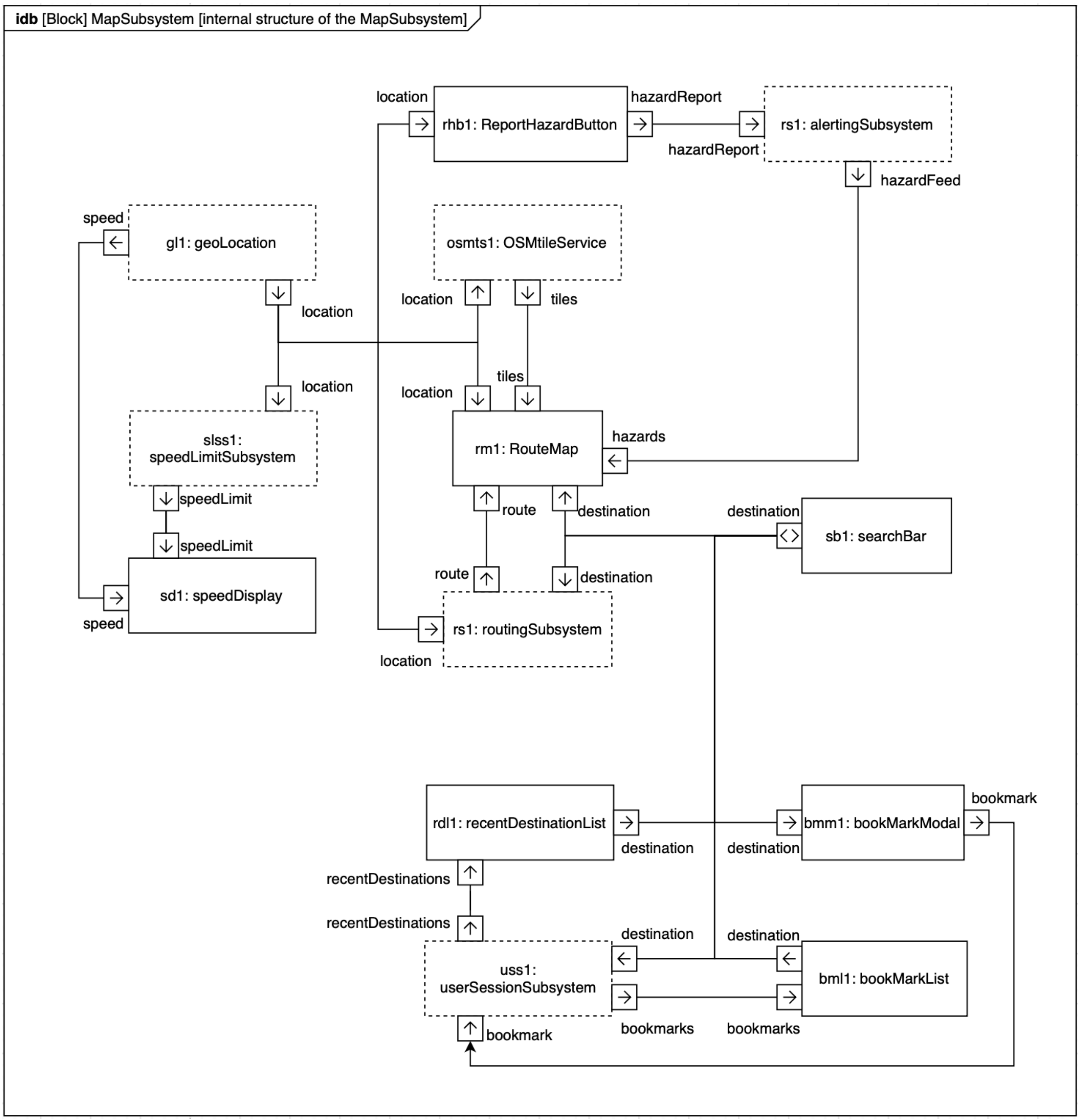


Figure 45: Internal Block Diagram: Map Subsystem

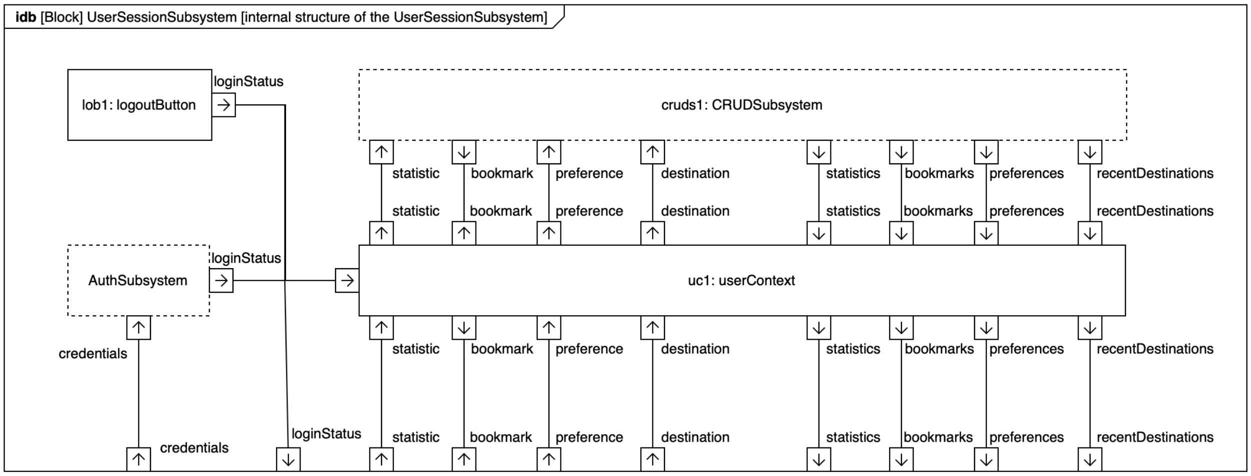


Figure 46: Internal Block Diagram: UserSession Subsystem

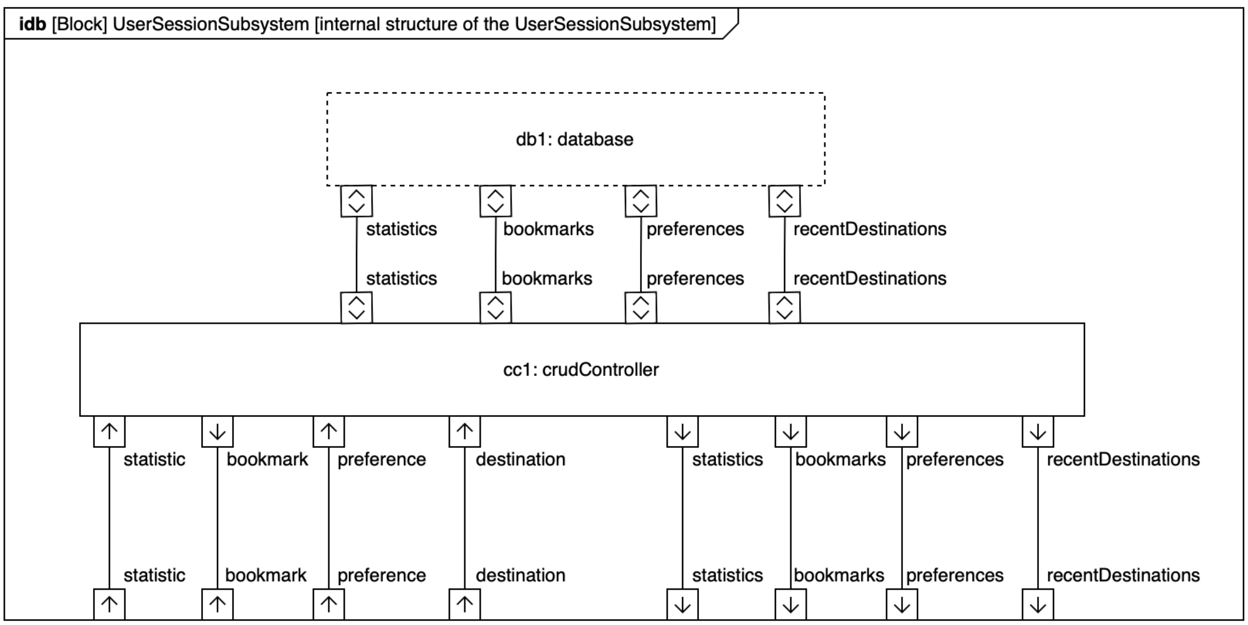


Figure 47: Internal Block Diagram: CRUD Subsystem

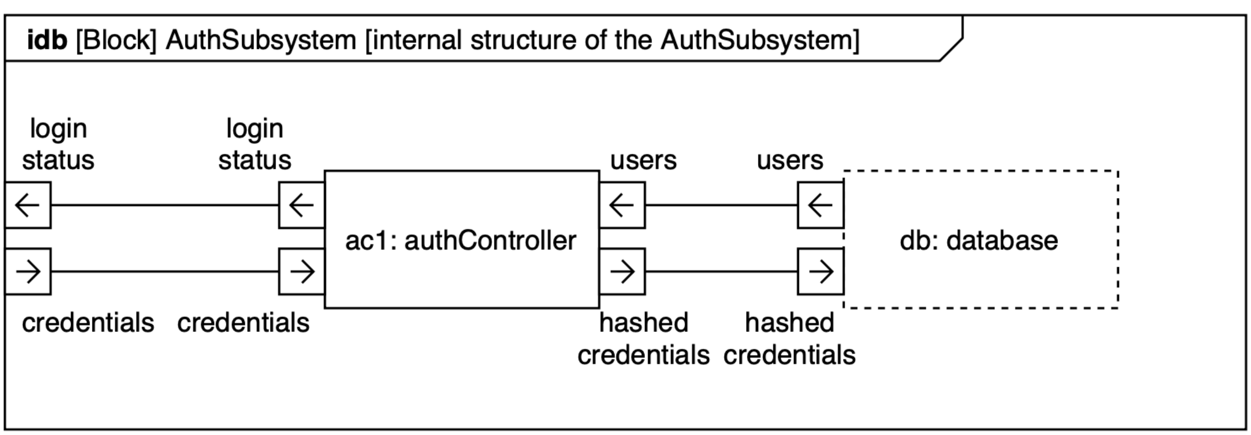


Figure 48: Internal Block Diagram: Auth Subsystem

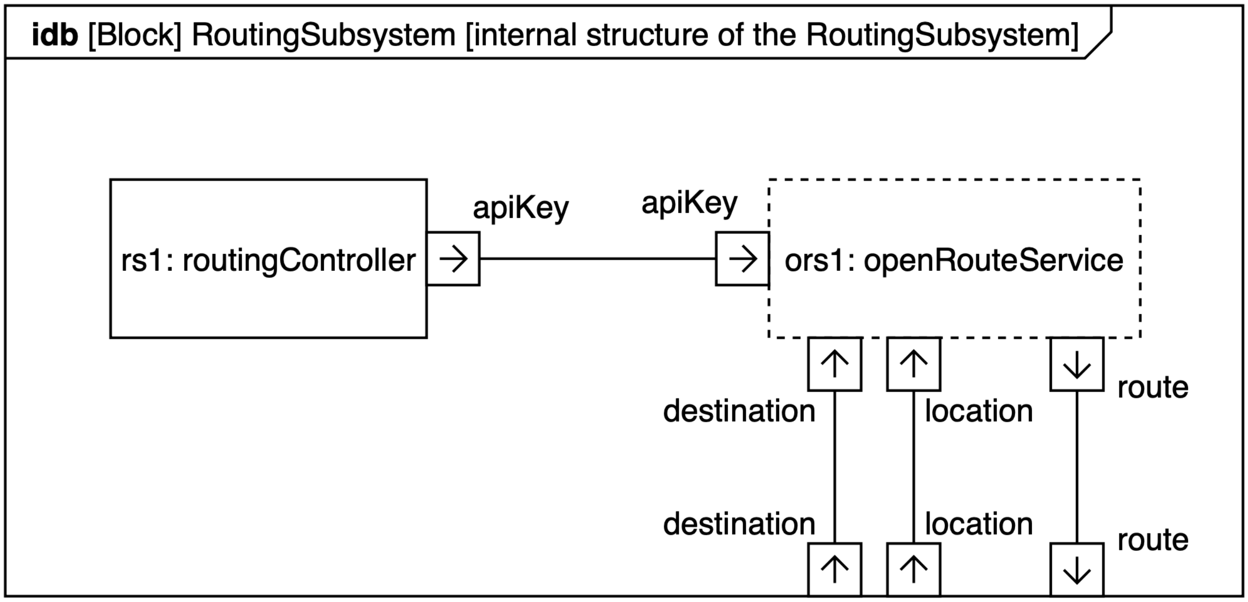


Figure 49: Internal Block Diagram: Routing Subsystem

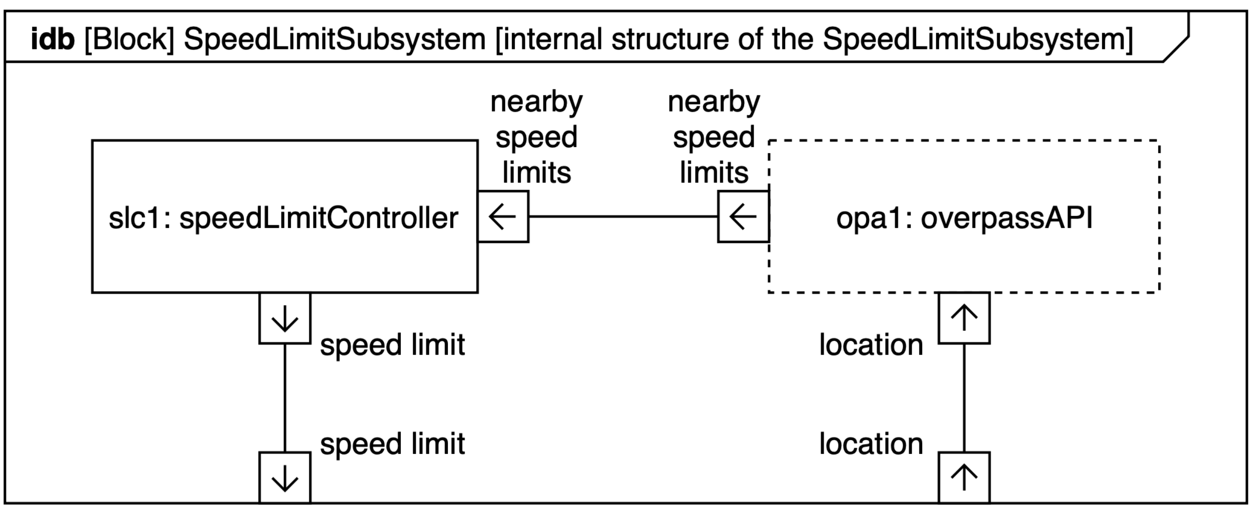


Figure 50: Internal Block Diagram: Speed Limit Subsystem

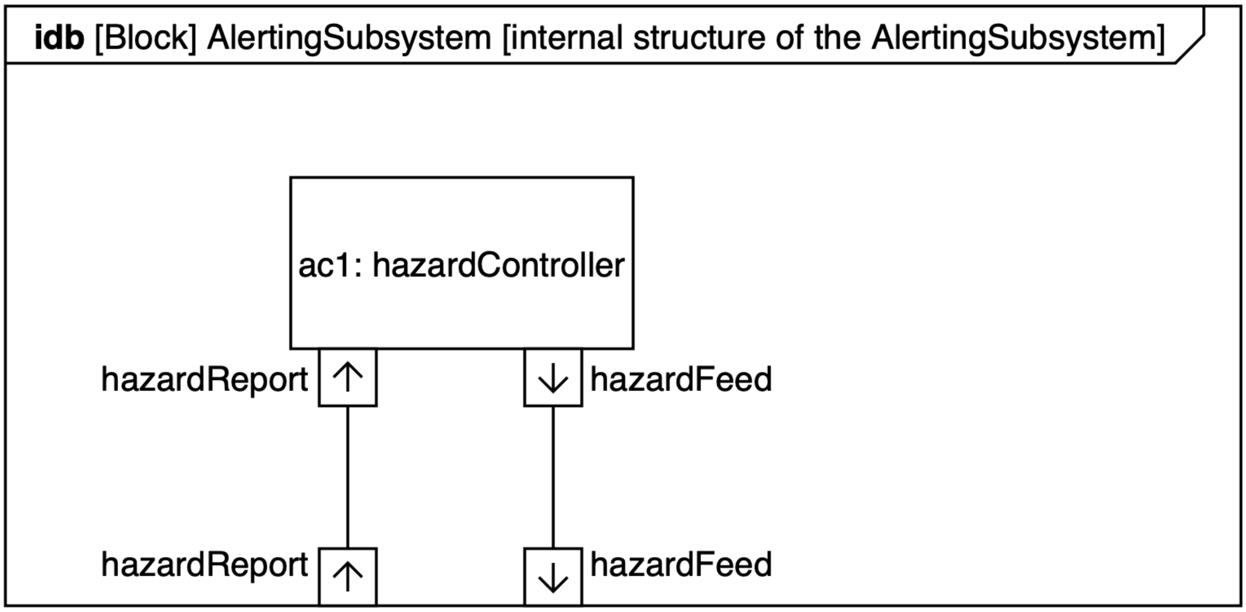


Figure 51: Internal Block Diagram: Alerting Subsystem

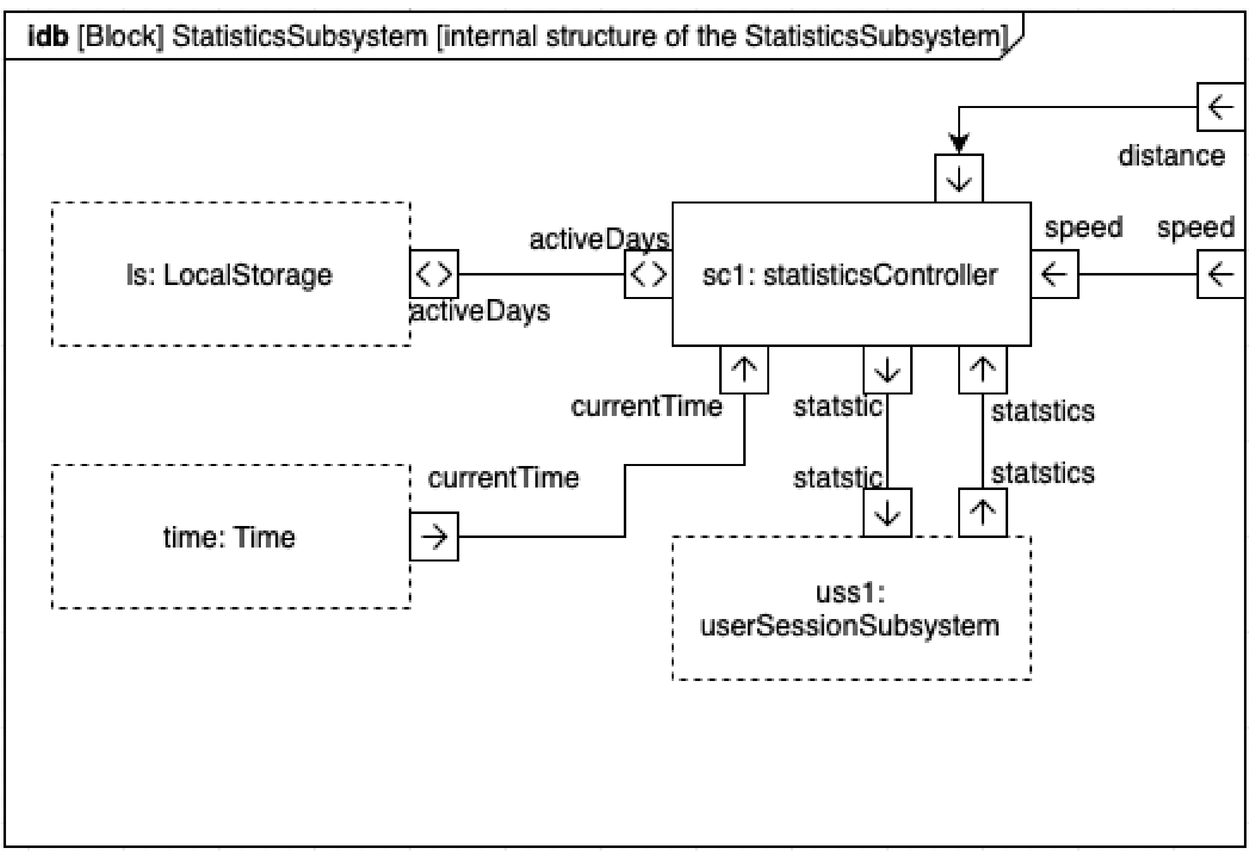


Figure 52: Internal Block Diagram: Statistics Subsystem

Deployment View

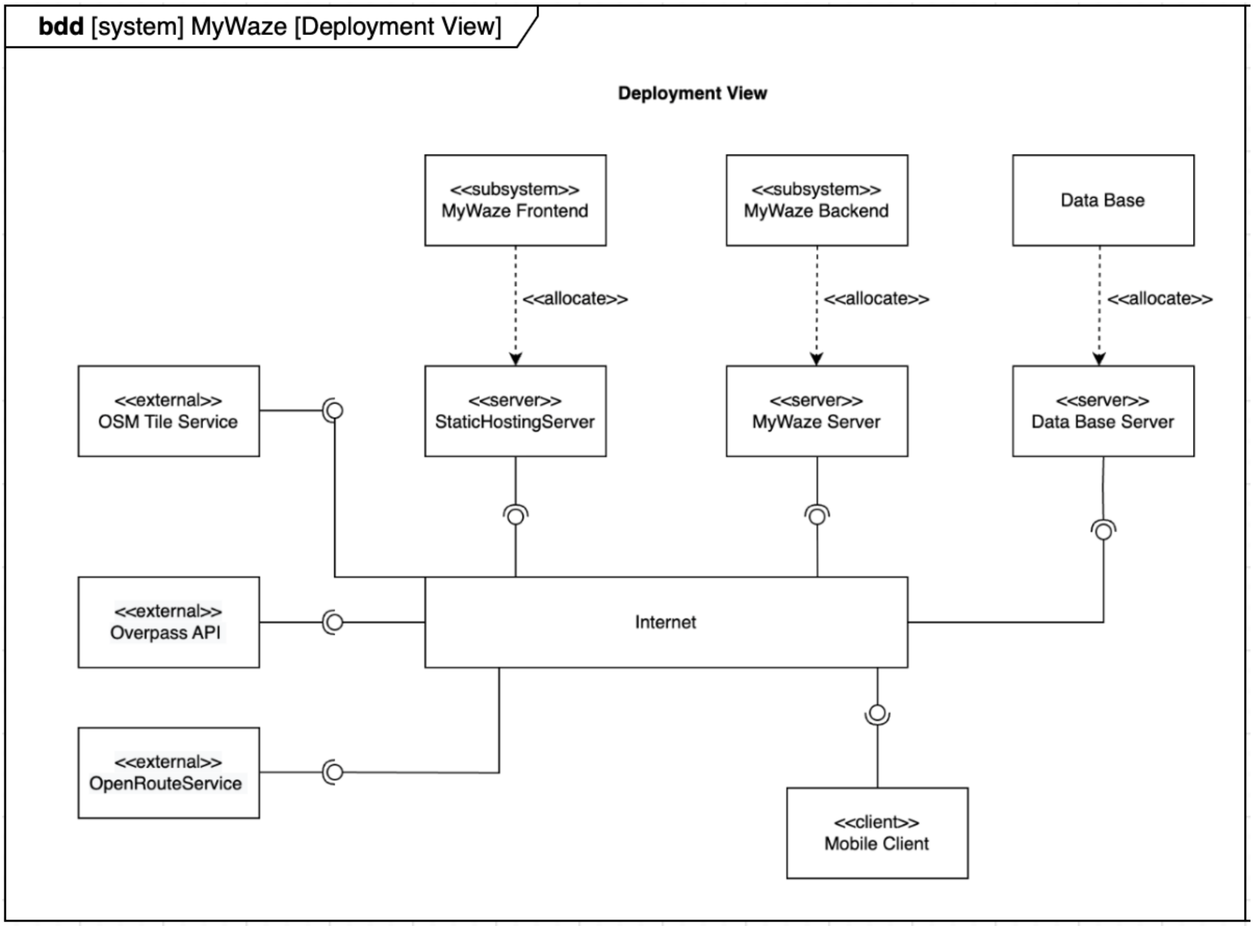


Figure 53: Deployment View

Variability Model

Feature Model

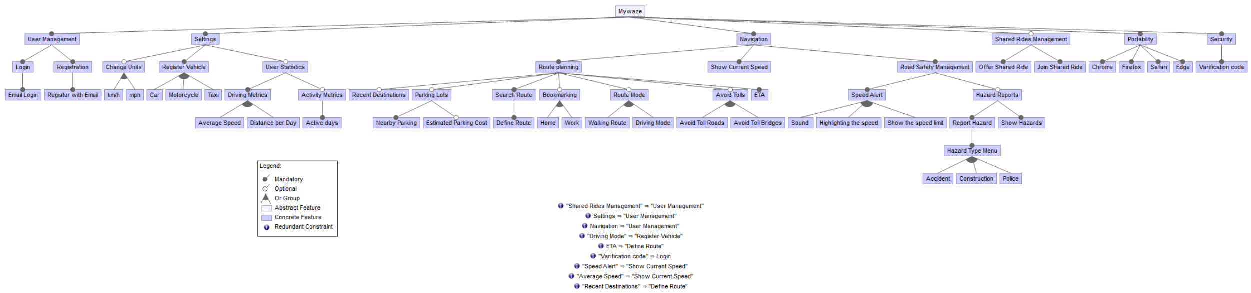


Figure 54: Feature Model

Configurations

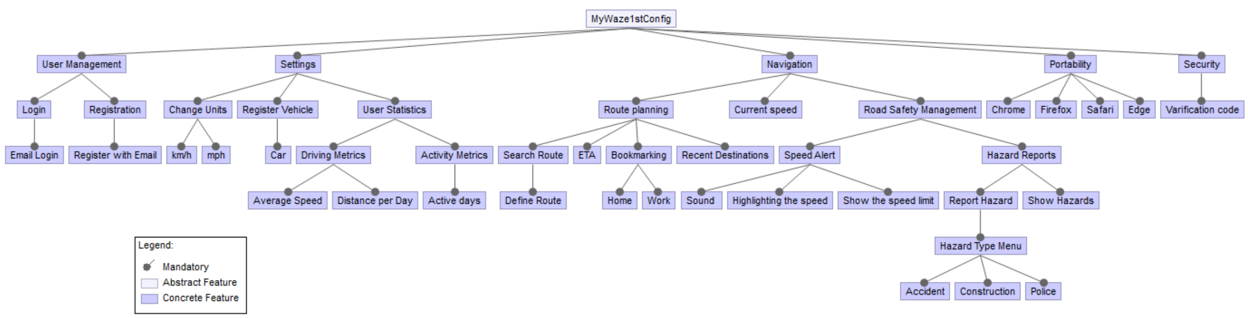


Figure 55: Configuration 1

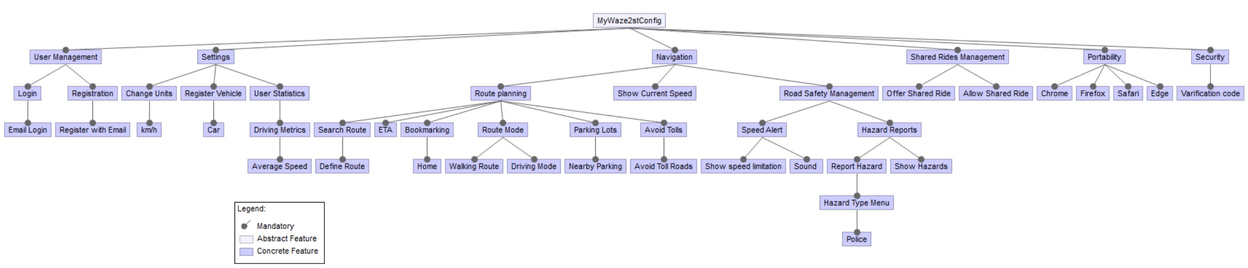


Figure 56: Configuration 2

Implementation Considerations for Non-Functional Requirements

Portability

The application is cross-browser compatible, functioning reliably on Chrome, Firefox, Safari, and other modern browsers. This is achieved through the use of React.js, which compiles to standard HTML, CSS, and JavaScript without requiring browser-specific dependencies.

Safety

To minimize driver distraction, the system supports hands-free interaction by allowing destination selection from bookmarks and recent destinations. Hazard reporting is facilitated via intuitive UI icons, enabling quick, non-intrusive input.

Reliability

The application demonstrates a fast recovery time (approximately 1 second) in the event of a crash or restart, aided by lightweight frameworks. It relies on stable, well-supported APIs, such as OpenStreetMap (OSM), for map data and route computation. A **layered architecture pattern** promotes maintainability and testability, simplifying issue detection and resolution.

Usability

The user interface is designed for accessibility and minimal interaction overhead. Key functions are located on the main map view, making them easily discoverable. Unit changes require only two clicks, and most actions are accessible with minimal user input, catering to a wide range of user demographics, especially regarding technological literacy.

Security

User accounts are secured through email verification and password protection, ensuring basic authentication and data protection measures are in place.

Performance

Route and ETA calculations are performed using efficient, third-party routing APIs, avoiding the overhead of custom algorithm implementations. The React.js-based single-page application leverages reusable components for maintainability and speed. Server-side computation ensures consistent performance regardless of client hardware (**client-server pattern**). The **Observer pattern** is employed for real-time hazard updates, eliminating the need for client-side polling.

Prototype

Running Instructions

Backend:

```
cd msp-mywaze-backend
npm i
npm run dev
```

Frontend:

```
cd msp-mywaze-frontend
npm i
npm run dev
```

The only prerequisites are a current version of `node` and `npm`. The database is implemented as SQLite, so no manual setup is necessary.

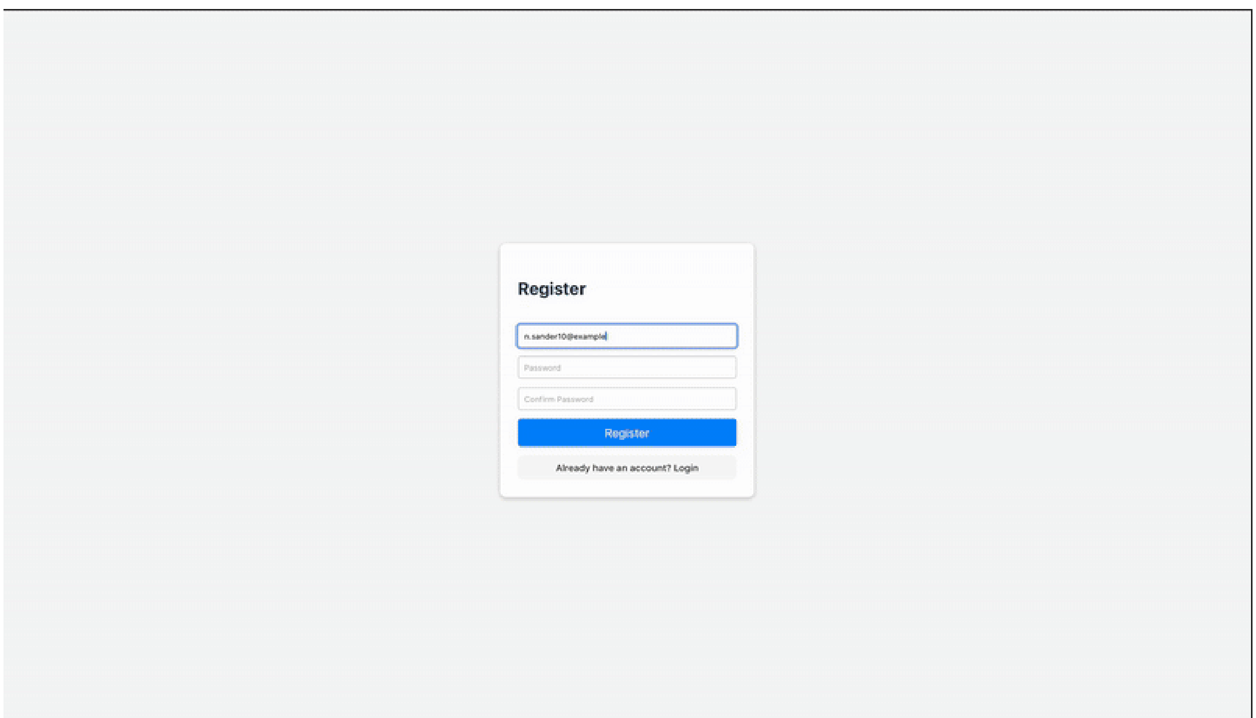
Routing uses OpenRouteService, which is quota-limited (daily) in the free version. If it stops working, please wait one day or change the variable `ORS_API_KEY` in `msp-mywaze-backend/src/routes/routing.ts` to your own (free) API key obtained from `https://openrouteservice.org`.

Features Implemented

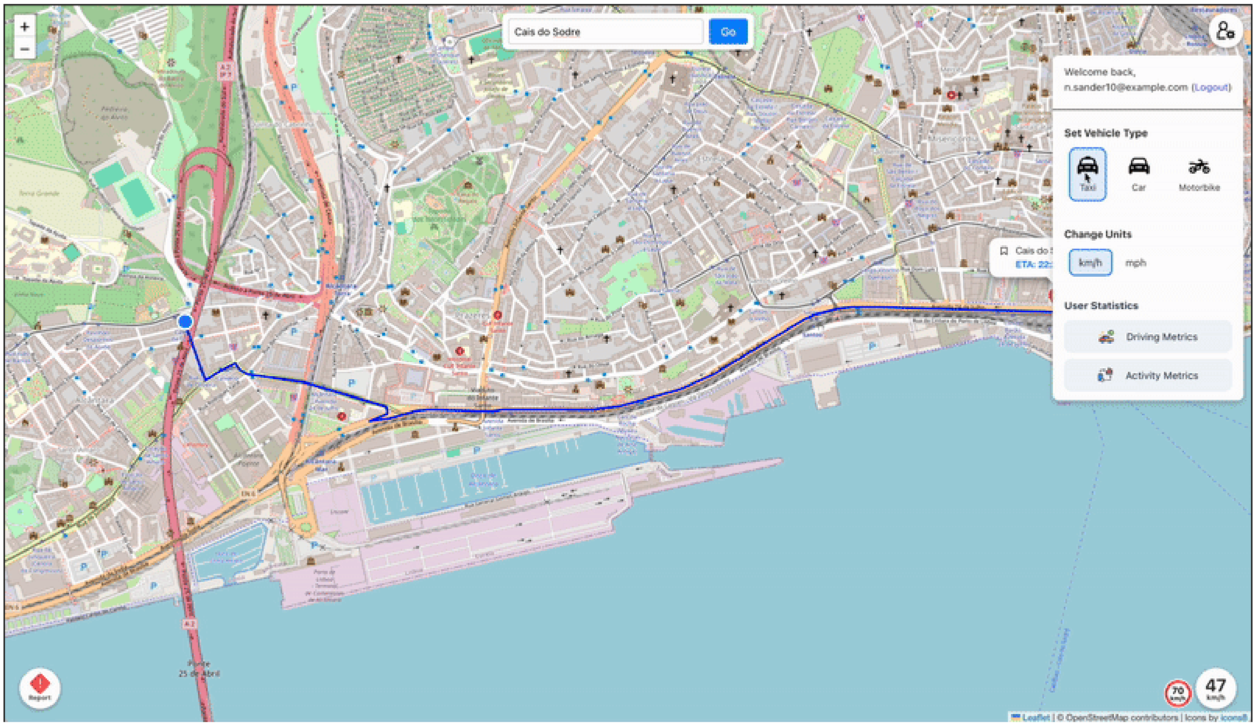
Video versions of the screenshots of all features can be found in the report/images/prototype directory, or in the readme file in the top level of the repository structure.

Basic Features

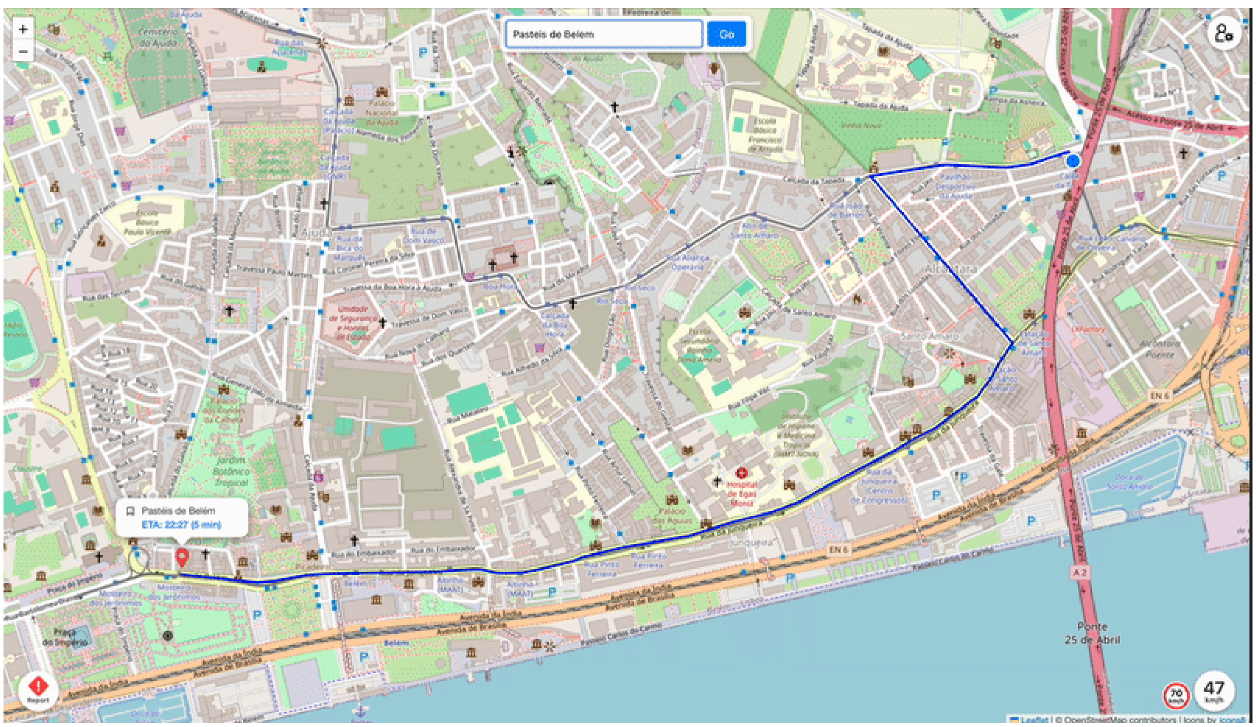
- **Registration**: User can register and login on the home page.



- **Vehicle Type Registration**: User can configure their vehicle type in the drop-down user menu on the top right of the screen.

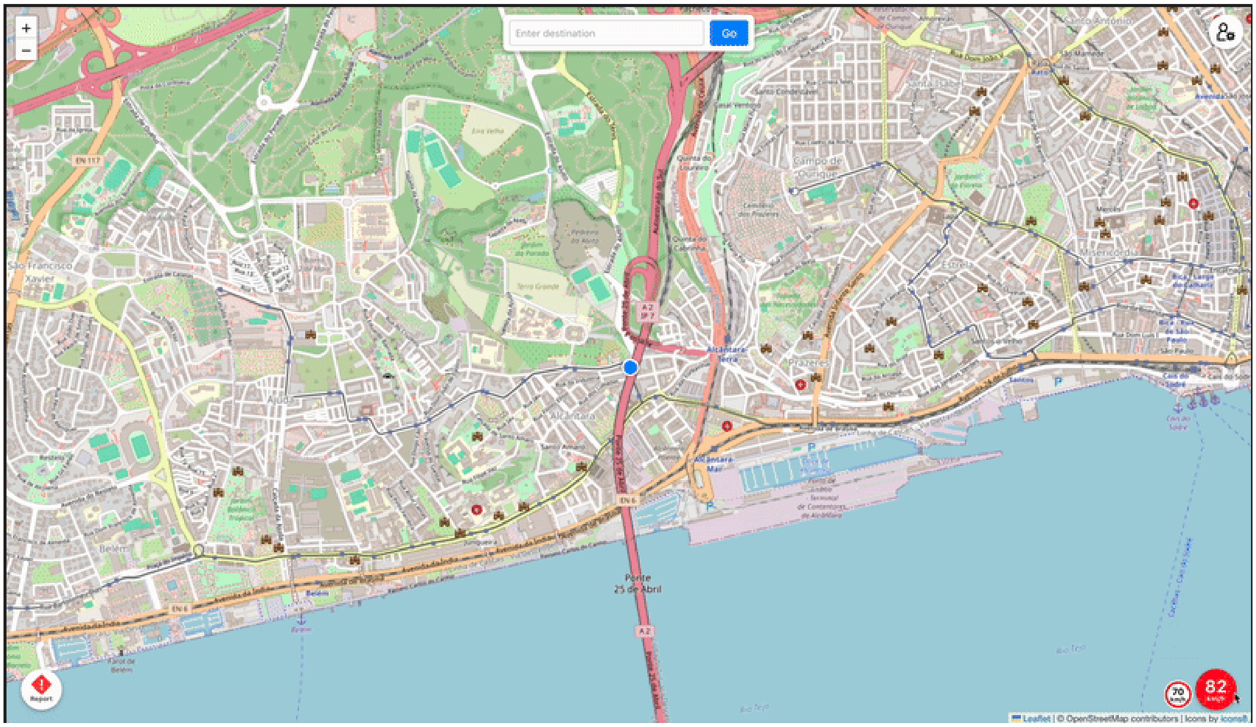


- **Define Route**: User can define a route by entering the destination name or address into the text field at the top center and clicking *Go*.



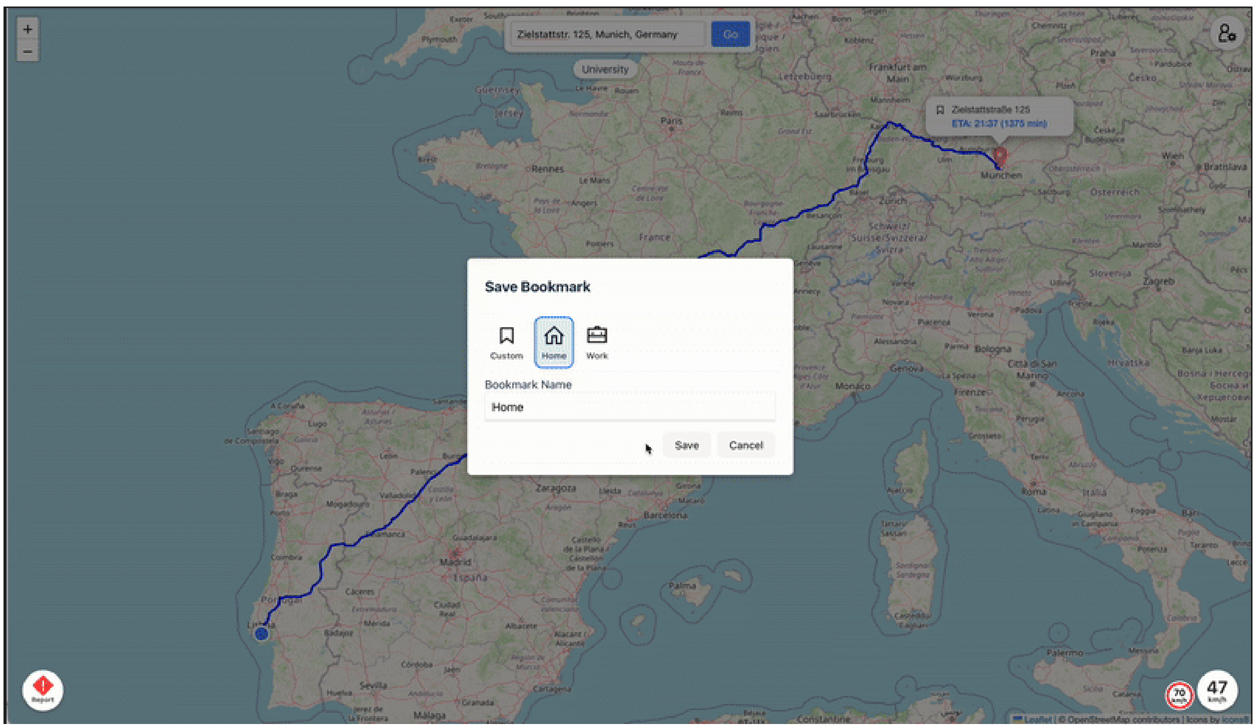
- **ETA**: ETA is displayed in a popup above the destination after a route is defined. (screenshot: see Define Route)

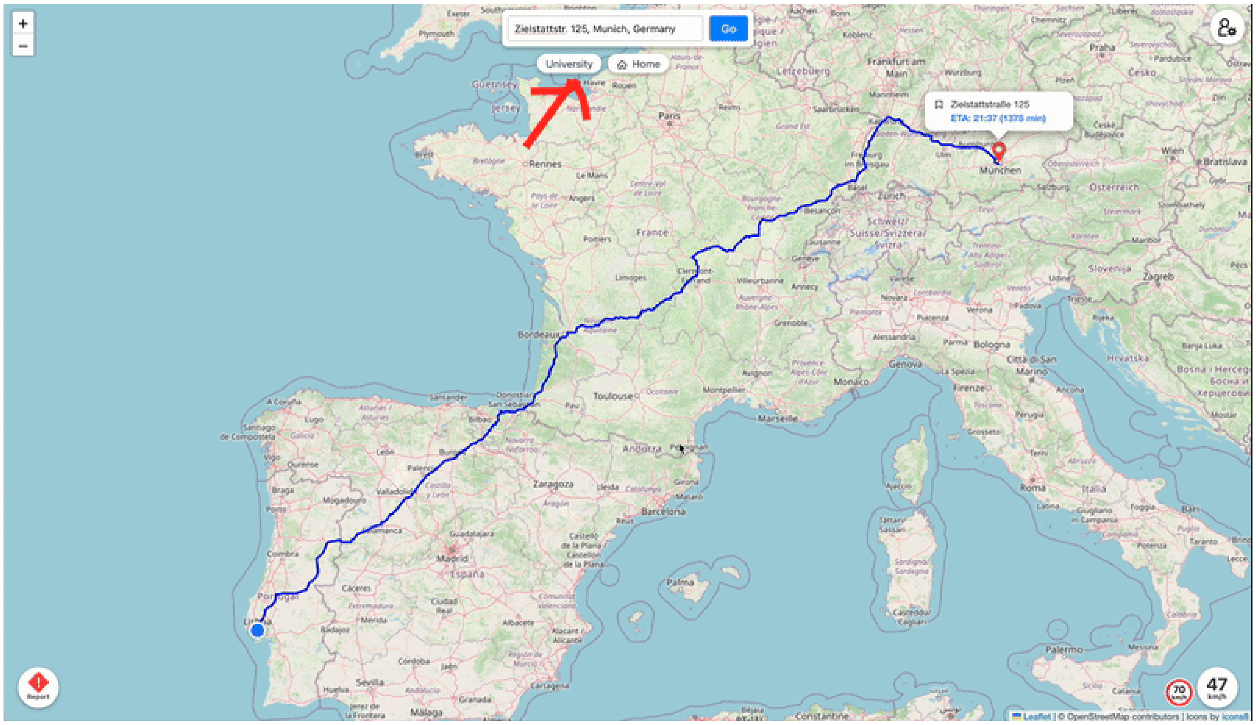
- **Speed Warning**: Current speed will turn red if the user moves over the speed limit. To test speed warnings on desktop, click on the speed value to generate a random speed around 50 km/h.



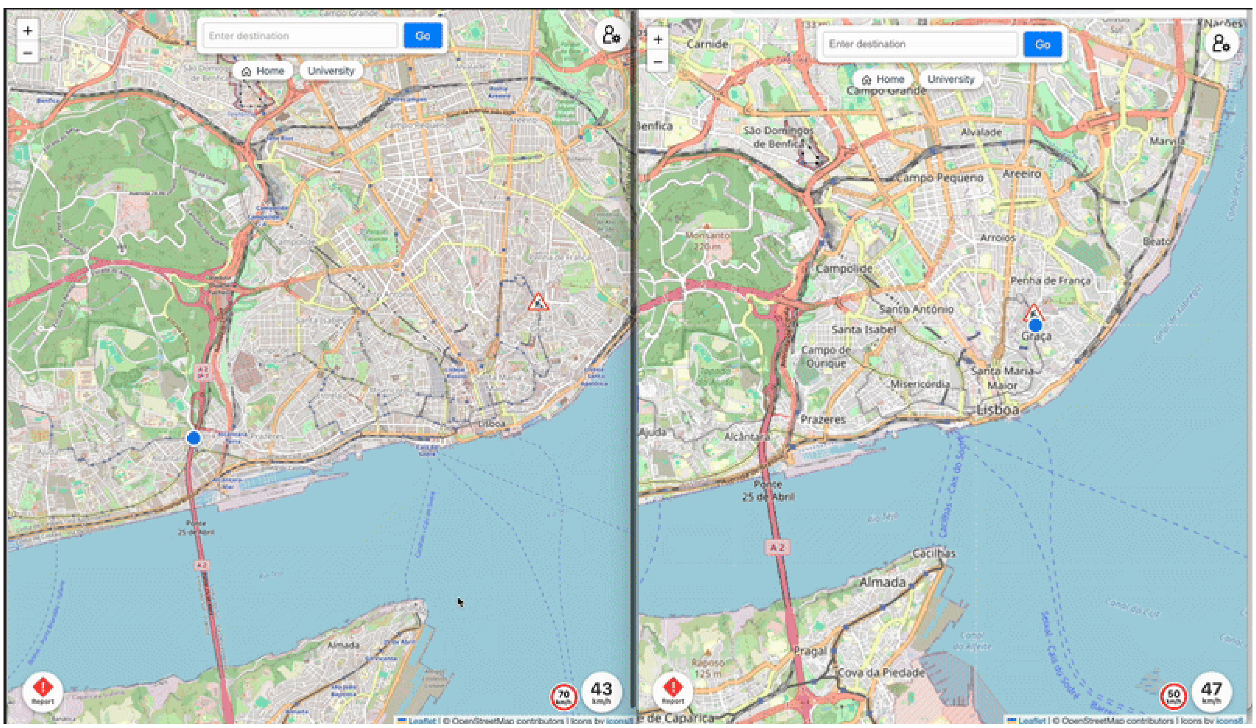
Advanced Existing Features

- **Bookmarking**: User can bookmark frequently used locations.

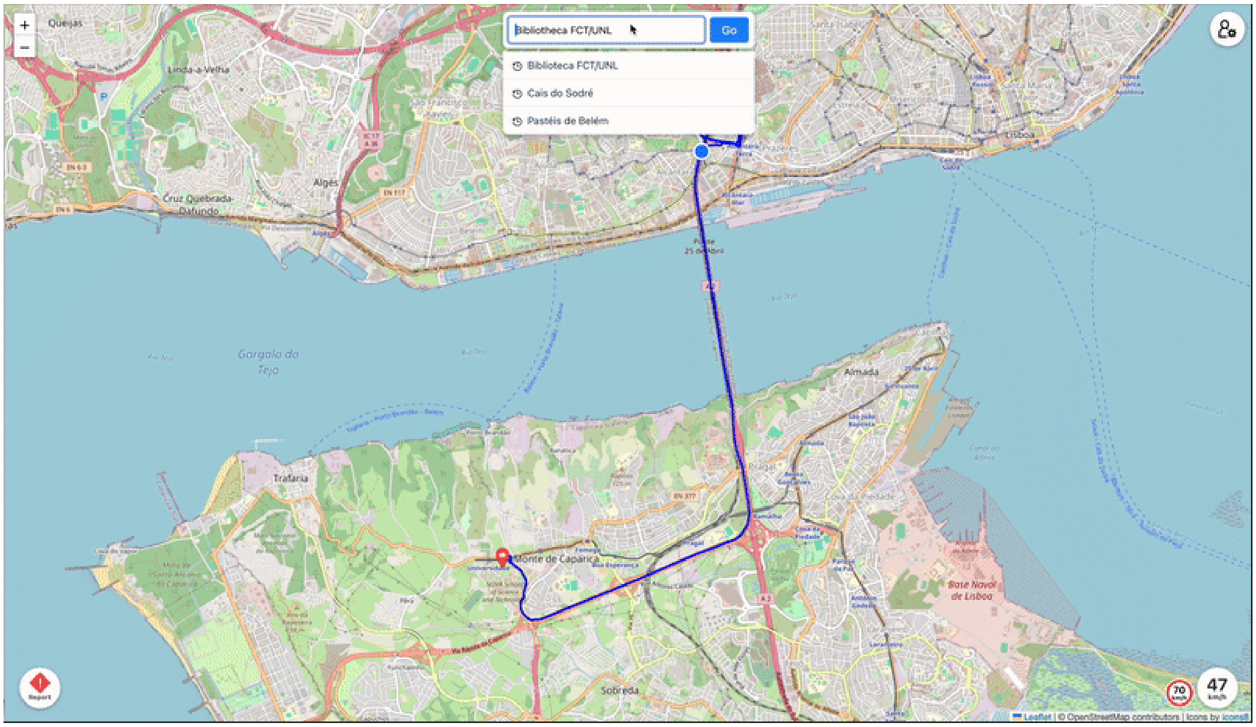




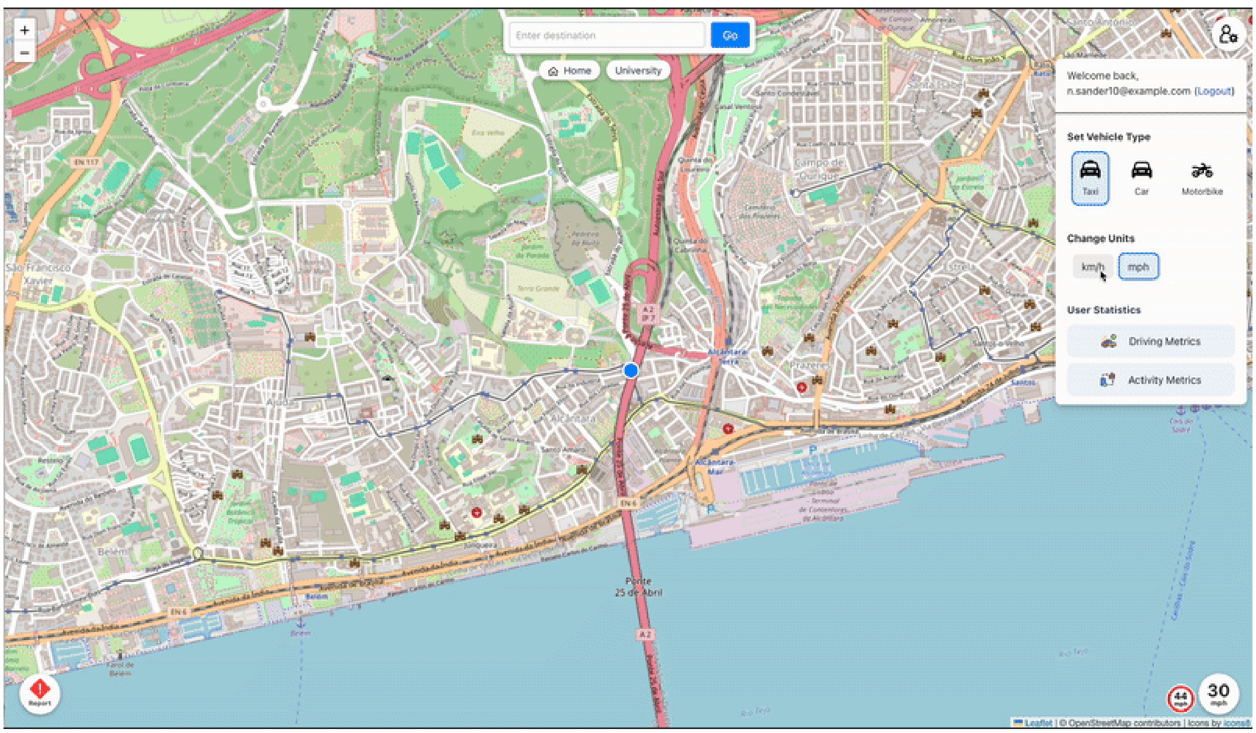
- **Hazard Reporting**: Users can report hazards encountered along the route. Reports are synchronized and shown across user sessions.



- **Recents**: Recently entered locations are shown for quick access.

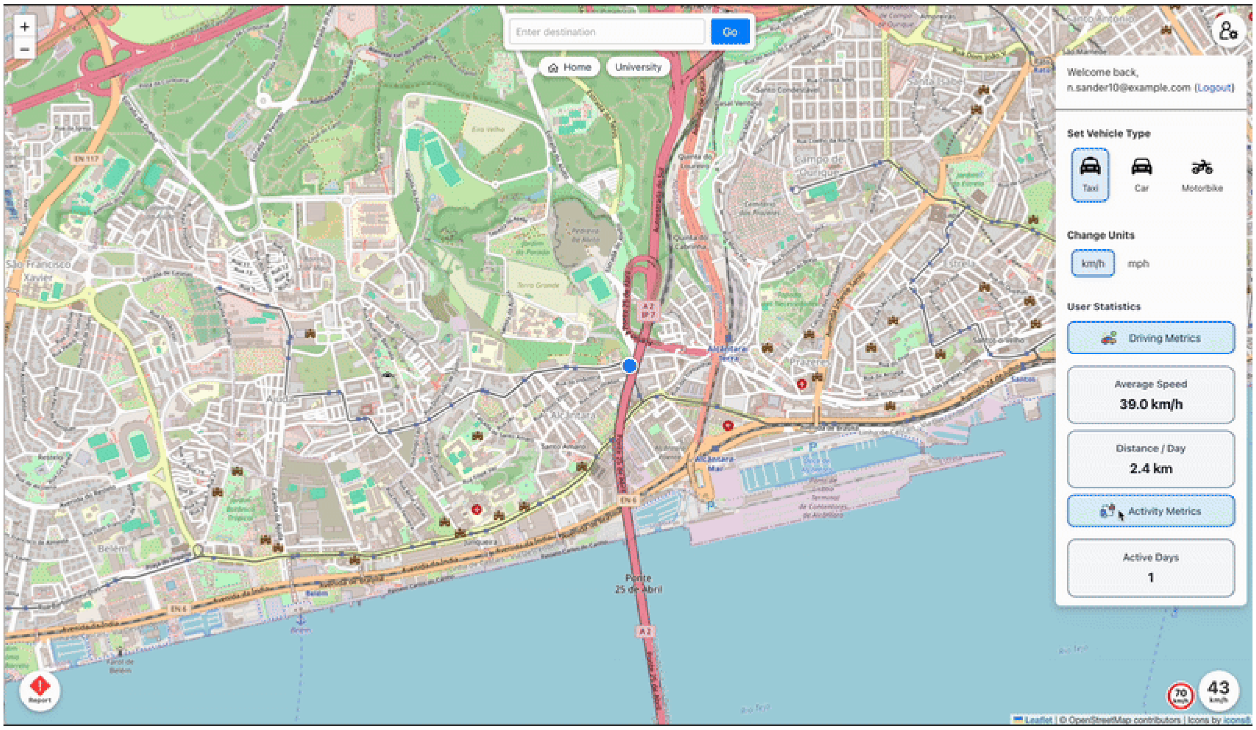


- **Units**: Users can switch between metric and imperial units.



Advanced New Features

- **Stats**: Displays statistics for average speed, distance, and active days.


Enter destination
Go
Home
University
Welcome back,
n.sander10@example.com (Logout)
Set Vehicle Type
Taxi
Car
Motorbike
Change Units
km/h
mph
User Statistics
Driving Metrics
Average Speed
39.0 km/h
Distance / Day
2.4 km
Activity Metrics
Active Days
1
Report
43
km/h
Leaflet | © OpenStreetMap contributors | Icons by icons8